

LLM-Powered Multi-Agent Negotiation and Trade Simulation: Analysing Negotiation Behaviour and Market Dynamics

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Abstract

This thesis investigates whether large language model (LLM) agents exhibit economically interpretable negotiation behaviour and whether coherent market-level dynamics emerge when many such agents interact under controlled bilateral bargaining conditions. A configurable multi-agent simulator serves as the experimental instrument: it implements buyer and seller agents that negotiate through alternating-offer price bargaining with private economic constraints, and logs every agent action at sufficient granularity for behavioural analysis. Six experiments (plus one extension) covering concession dynamics, deadline pressure, market price discovery, shock response, supply–demand shifts, informational anchor environments, and matching mechanism design were conducted across 3,420 sessions using `Qwen/Qwen2.5-14B-Instruct` with free-language agents and the feasibility gate disabled, so that every acceptance is LLM-driven.

At the negotiation level, LLM agents produce concave (Boulware-style) concession patterns with a 5.65:1 buyer-to-seller concession ratio. These results indicate that AI sellers maintain high price rigidity, sticking firmly to their initial anchors compared to their buyer counterparts. Deadline pressure accelerates settlement: 82.4% of deals close in the final two rounds under a six-round horizon.

At the market level, prices exhibit strong stickiness in response to structural shocks (demand elasticity 0.275; supply elasticity 0.035). An informational anchor environment experiment demonstrates that the reference price signal in the agent prompt is a primary determinant of price levels, with mean transaction prices spanning approximately \$30 across anchor modes under identical fundamentals. Surplus-maximising matching improves total welfare by 7.2% relative to random matching. Taken together, these results reveal a micro–macro capability gap: agents close deals effectively at the session level yet fail to produce reliable price discovery in the aggregate.

Several behavioural patterns—concave concession trajectories, deadline-driven settlement acceleration, and reference-price sensitivity—are directionally consistent with patterns documented in the human negotiation literature, suggesting that LLM agents can serve as credible proxies for

studying bargaining dynamics in controlled settings. The results establish a configurable experimental platform connecting micro-level agent behaviour to macro-level market outcomes, and lay groundwork for future work on prompt design, multi-model comparison, and institutional mechanism optimisation.

1 Introduction

1.1 Background

Large language models are increasingly deployed as autonomous agents that interact with environments, tools, and other agents. Recent work has shown that LLM-powered agents can exhibit emergent social behaviours in open-ended simulation [10], coordinate within multi-agent frameworks [19], and interleave reasoning with action [20]. A natural extension of this research agenda is to ask whether LLM agents can engage in structured economic interaction: bilateral bargaining under private constraints, with real stakes, limited information, and measurable outcomes. Open-ended social simulation demonstrates that LLM agents can sustain coherent dialogue and exhibit personality-consistent behaviour, but it does not require agents to satisfy hard economic feasibility constraints, discover prices, or produce outcomes that can be evaluated against classical economic benchmarks. Structured economic interaction imposes a qualitatively different test.

Bilateral bargaining is among the best-characterised settings in economic theory. Rubinstein’s alternating-offer model [14] predicts that patient agents extract larger surplus shares through time-preference discounting. The anchoring literature [16] establishes anchoring as a general cognitive heuristic; in negotiation contexts, Galinsky and Mussweiler [2] show that first offers systematically bias final agreements. Deadline pressure effects have been documented in human experiments [13], and decentralised bilateral markets are expected to produce equilibrium prices with persistent price dispersion [15]. These theoretical predictions provide reference points against which observed agent behaviour can be compared directionally, though the extent to which any simulator-based evaluation constitutes a clean test depends on the prompt and protocol design.

Despite growing interest in LLM negotiation, existing studies typically focus on multi-issue dialog games [7, 1], personality-driven interaction [9], or assistive human–AI bargaining [5]. These designs evaluate negotiation capacity, but do not systematically test LLM agents against the specific quantitative predictions of classical bargaining theory, nor do they examine whether LLM agents reproduce qualitative behavioural patterns documented in human negotiation studies – such as deadline-driven concession acceleration [13], Boulware-style tactical hardening, or first-offer anchoring [2]. Furthermore, most studies operate at a single level of analysis without connecting micro-level bargaining outcomes to macro-level market dynamics within a single controlled platform. As automated negotiation expands into consumer and enterprise markets [22], understanding these behavioral gaps becomes increasingly critical. These developments mo-

tivate a systematic, simulator-based approach to evaluating how LLM agents navigate complex bargaining environments.

1.2 Problem Statement

The central question motivating this thesis is:

What patterns of concession, price discovery, and market efficiency emerge when LLM agents negotiate under controlled bilateral bargaining conditions, and to what extent are observed outcomes shaped by the informational and protocol environment in which agents operate?

Understanding these behaviors requires looking beyond LLM outputs in isolation. Instead, the entire simulator’s design—from the prompt architecture shaping agent reasoning to the interaction protocols governing turns and termination—critically determines the observable results. These design choices must be made explicit and independently configurable to enable defensible interpretation of experimental findings. A further challenge is attribution: any observed outcome in an LLM simulation reflects the combined system of model, prompt, and protocol. Disentangling the contribution of LLM decision-making from the contribution of simulator infrastructure requires careful logging, explicit ablation studies, and conservative interpretation. This thesis builds a platform designed to support such analysis, served as the instrument; the primary subject of investigation is the behaviour of LLM agents within it.

1.3 Research Objectives

A configurable multi-agent negotiation simulator was designed and implemented to support the empirical programme described below. The system enforces private economic constraints, supports a structured alternating-offers protocol, and provides event-level logging sufficient for session and market-level analysis. The research objectives are:

1. Characterise LLM bargaining behaviour empirically—concession dynamics, deadline sensitivity, and deal success—and compare observed patterns directionally against predictions from classical bargaining theory.
2. Investigates how LLM-agent markets respond to shifting fundamentals—such as supply-demand fluctuations and external shocks—as well as informational influences like reference-price anchoring. Furthermore, it assesses whether reliable price discovery emerges from bilateral negotiations under these controlled conditions
3. Evaluate whether institutional mechanism design (buyer-seller matching algorithms) can improve market-level welfare independently of agent’s own negotiation capabilities.

4. Identify how simulator design choices—prompt architecture, feasibility enforcement, and reference-price framing—function as protocol-level design considerations, and report methodological cautions relevant to LLM-based economic simulation.

1.4 Contributions

This thesis makes four contributions.

1. Empirical characterisation of LLM bargaining behaviour. LLM agents produce concave (Boulware-style) concession trajectories with a 5.65:1 buyer-to-seller concession ratio, and deadline pressure concentrates settlement in the final rounds (82.4% of deals close in the final two rounds under a six-round horizon). Price stickiness is substantial: demand and supply elasticities of 0.275 and 0.035 respectively indicate that agents update prices only weakly in response to changing fundamentals. Several of these patterns are directionally consistent with behavioural patterns documented in the human negotiation literature, though no direct human comparison was conducted and the underlying mechanisms may differ.

2. Demonstration that prompt-embedded information is a first-order determinant of market outcomes. An anchor-environment experiment (Experiment 4 extension) shows that the reference-price signal in the agent prompt produces a mean transaction price spread of approximately \$30 across anchor modes under identical fundamentals, and reverses the sign of shock effects between anchor conditions. This establishes prompt-level information framing as a first-order experimental variable in LLM-based market simulation, with direct implications for the interpretation of all other experiments.

3. Identification of a micro-macro capability gap. LLM agents function as effective instruments for deal-closing—demonstrating a 78.6% overall success rate and 81% market liquidity—yet remain ineffective at driving price discovery. In these simulations, prices exhibit persistent dispersion without convergence, and elasticities remain near zero. This disconnection between individual agreement-seeking and aggregate price efficiency identifies two distinct dimensions of negotiation competence that are often conflated in existing evaluation frameworks.

4. A configurable experimental platform connecting micro-level bargaining to macro-level market dynamics, with hard private constraints, structured event logging, and explicit controls over design choices including gate status, prompt architecture, reference-price mode, and matching algorithm. The platform enables the findings above and provides a foundation for future ablation and extension.

The remainder of this thesis is organised as follows. Section 2 reviews related work in LLM-based simulation, LLM negotiation, and classical bargaining theory. Section 3 describes the problem formulation and simulator architecture. Section 5 details the methodology and experimental design for all experiments. Section 6 presents results and analysis. Section 7 discusses implications, limitations, and the role of simulator design choices as experimental variables.

Section 8 concludes and identifies directions for future work.

2 Literature Review

This work sits at the intersection of agent-based simulation, LLM-based agent design, and computational bargaining research. We survey these threads and identify the gap that motivates the experimental programme.

2.1 Agent-Based Modeling and LLM-Based Simulation

Agent-based modelling (ABM) has long been used to study emergent economic phenomena from micro-level interactions, including market price formation, information diffusion, and collective behaviour [3]. Traditional ABM agents follow scripted rules or simple utility functions, which limits behavioural realism. The advent of large language models has opened a new direction: using LLMs as behavioural engines, so that agents respond to context in natural language and exhibit richer, less predictable strategies.

Several systems have demonstrated this potential at scale. S3 [4] simulates social-network dynamics with LLM agents, studying opinion formation and information spread. AgentSociety [11] scales LLM-driven agents to large populations, capturing emergent social and economic phenomena. Williams and Hosseinmardi [18] apply generative agents to epidemic modelling, showing that LLM agents can reproduce stylised public-health dynamics in structured environments. Gao et al. [3] survey the broader LLM-ABM landscape and identify economic simulation as a high-potential but under-explored domain. This thesis contributes to this domain by systematically testing LLM agents against classical bargaining predictions within a bilateral negotiation protocol under hard economic constraints, connecting micro-level agent behaviour to macro-level market outcomes.

2.2 Generative Agents and Agent Architecture

Park et al. [10] introduced *Generative Agents*, demonstrating that LLM-powered characters in a sandbox environment exhibit emergent social behaviours—forming relationships, spreading information, and coordinating activities—without explicit scripting. This established LLMs as plausible behavioural engines for simulation, though the focus was on open-ended social dynamics rather than constrained economic decision-making.

At the architectural level, the ReAct framework [20] introduced a reasoning-and-acting paradigm in which models interleave chain-of-thought reasoning with environment actions. Multi-agent orchestration frameworks such as AutoGen [19] provide general infrastructure for LLM agent interaction but target open-ended tool use rather than controlled economic experiments. This thesis embeds LLM agents within a fixed alternating-offer protocol with hard feasibility constraints,

enabling quantitative comparison against theoretical baselines—a form of evaluation that open-ended simulation frameworks are not designed to support.

2.3 LLM Negotiation Literature

Negotiation is among the earliest testbeds for learned agents. Lewis et al. [7] trained neural negotiation agents via reinforcement learning and self-play on a multi-issue bargaining task, showing that learned models reach agreements but can degenerate into non-human-like strategies. That work predates LLM-based approaches and uses task-specific training; the agents in this thesis are prompted, not trained, and negotiate a single price under private constraints rather than trading across multiple issues.

Classical bargaining theory provides the normative benchmarks against which LLM behaviour is evaluated throughout this paper. Rubinstein’s [14] alternating-offer model predicts that a more patient agent extracts a larger share of surplus; the experiments test whether LLM agents reproduce this effect. The anchoring literature, originating with Tversky and Kahneman [16] and refined in negotiation contexts by Galinsky and Mussweiler [2], predicts that first offers bias final agreements. Roth, Murnighan, and Schoumaker [13] established that human bargainers make disproportionately large concessions near deadlines; the experiments test this prediction by varying time horizons. At the market level, Rubinstein and Wolinsky [15] showed that decentralised bilateral bargaining produces persistent price dispersion, predicting that the Law of One Price fails under bilateral matching. Raiffa’s [12] concept of the zone of possible agreement (ZOPA) is central to the feasibility gate and to the observation that ZOPA width strongly predicts deal success in the experiments. On the computational side, Zheng et al. [21] demonstrate that multi-agent reinforcement learning can discover effective economic policies in simulated economies, motivating the use of AI agents for mechanism-design experiments.

With the advent of large language models, several works have studied LLM behaviour in richer negotiation settings. Abdelnabi et al. [1] evaluate LLMs in interactive multi-agent negotiation games, finding that deliberation-style prompting improves agreement quality; their multi-issue, multi-party design complements the single-issue bilateral setting used here, which isolates price dynamics from issue-trading effects. Mao et al. [9] show that personality prompts measurably shift negotiation outcomes, motivating a planned communication-strategy experiment (deferred to future work). Chawla et al. [5] and Amato et al. [6] examine LLM-based negotiators in cooperative and game-theoretic settings respectively, while Davidson et al. [22] model agent-to-agent negotiations in consumer markets; collectively, these works explore richer environments than the bilateral single-price protocol used here, but none systematically control for the role of simulator design in shaping observed behaviour. The chain-of-thought prompting literature [17] underpins the deliberative reasoning pattern used in the agent prompts.

2.4 Research Gap

The literature reviewed above leaves four gaps that motivate this work.

First, existing LLM simulation platforms operate primarily at either the micro level—studying individual agent behaviour and reasoning—or the macro level—studying aggregate social or economic outcomes—but no single platform systematically connects both within a controlled experimental design.

Second, while classical bargaining theory provides precise quantitative predictions, no prior work has tested these predictions against LLM agents in a systematic, reproducible manner; most studies report qualitative or task-specific findings rather than comparing against theoretical benchmarks such as Rubinstein equilibrium splits or deadline-effect magnitudes.

Third, and most critically, the extent to which simulator and protocol design choices are first-order determinants of observed LLM negotiation behaviour has not been adequately recognised in prior work. Prompt formulation, reference-price framing, and feasibility enforcement can each substantially shift the apparent behaviour of agents; yet these choices are rarely reported in detail or systematically controlled. This raises a validity question that applies to a wide range of existing findings: how much of what is reported as LLM negotiation behaviour in prior work is attributable to genuine agent cognition, and how much is induced by underspecified experimental choices?

Fourth, no prior work has systematically examined whether LLM agents reproduce the qualitative behavioural patterns documented in human negotiation studies—such as deadline-driven concession acceleration [13], Boulware-style tactical hardening [12], and first-offer anchoring [2]—within a controlled bargaining protocol that enables direct comparison against those benchmarks.

The experimental programme reported in this thesis addresses these gaps by constructing a single configurable platform that connects micro-level bargaining dynamics to macro-level market outcomes, provides quantitative comparisons against classical predictions across six experiments (plus one extension), and makes simulator design choices—feasibility gating, prompt formulation, communication strategy, matching mechanism—explicit and selectively ablated. The contribution is methodological and empirical rather than algorithmic: the platform offers a reproducible testbed with hard economic constraints and deterministic fallbacks, not a new model architecture or learning algorithm.

The literature reviewed above leaves four distinct gaps that motivate this work:

- 1. The Level-of-Analysis Gap:** Existing LLM simulation platforms operate primarily at either the *micro level*—studying individual agent behaviour and reasoning—or the *macro level*—studying aggregate social or economic outcomes. No single platform systematically connects both within a unified experimental design, leaving the causal link between individual bargaining behaviour and emergent market dynamics unexplored.
- 2. The Theoretical Benchmark Gap:** While classical bargaining theory provides precise quantitative predictions, prior work has seldom tested LLM

agents against these formal benchmarks in a reproducible manner. Most studies report qualitative or task-specific findings rather than evaluating LLM performance against rigorous theoretical foundations, such as Rubinstein equilibrium splits or deadline-effect magnitudes.

3. **The Methodological Validity Gap:** The extent to which simulator and protocol design choices function as *first-order determinants* of behaviour has not been adequately recognised. Factors such as prompt formulation, reference-price framing, and feasibility enforcement can substantially shift an agent’s apparent performance. This raises a critical validity question: how much of the reported behaviour is attributable to genuine agent cognition versus the artefacts of an underspecified experimental environment?
4. **The Behavioral-Alignment Gap:** No prior work has systematically examined whether LLM agents reproduce the *qualitative behavioural patterns* documented in human negotiation literature—such as deadline-driven concession acceleration [13], Boulware-style tactical hardening [12], and first-offer anchoring [2]—within a controlled protocol that enables direct, empirical comparison against human benchmarks.

The experimental programme reported in this thesis addresses these gaps by constructing a single, configurable platform that connects micro-level bargaining dynamics to macro-level market outcomes. It provides quantitative comparisons against classical predictions across six experiments (plus one extension) while making simulator design choices—specifically feasibility gating, prompt formulation, communication strategy, and matching mechanisms—explicit and selectively ablated. The contribution is fundamentally **methodological and empirical**: the platform establishes a reproducible testbed that enforces hard economic constraints and utilizes deterministic fallbacks to evaluate LLM bargaining and negotiating behaviour.

3 Problem Formulation

We formalise the bilateral bargaining setting instantiated by the simulator and define the outcome quantities measured across all experiments.

3.1 Buyer–Seller Negotiation Setting

The simulator implements a two-player alternating-offer protocol over a single commodity with a bounded round horizon, using an alternating-offer turn structure as in Rubinstein’s bargaining model [14], but with a finite horizon and hard private constraints.

In each negotiation session, one buyer and one seller exchange priced offers over at most R rounds (configurable; default $R=10$). The buyer moves first at round 0; agents alternate on subsequent rounds. On each turn, the active agent

selects an action from $\{\text{OFFER}, \text{COUNTER}, \text{ACCEPT}, \text{REJECT}\}$ with an associated price (for OFFER and COUNTER) or null (for ACCEPT and REJECT).

A session terminates under one of three conditions:

- **Accepted:** an agent outputs ACCEPT, and the deal settles at the opponent’s last offered price.
- **Rejected:** an agent outputs REJECT, ending the negotiation with no deal.
- **Timeout:** the round limit R is reached without agreement.

The buyer-first turn order gives the buyer a structural first-mover advantage. Its effect on surplus distribution at the micro level is documented in Section 6; the effect does not persist at the aggregate market level (Finding 8).

3.2 Private Constraints

Each agent holds private constraints that are never shared with the opponent and are enforced deterministically by the simulator.

Buyer. A buyer is characterised by a value v_b (maximum willingness-to-pay for the item) and a budget B_b (hard spending cap). The effective price ceiling is

$$\bar{p}_b = \min(v_b, B_b).$$

An offer at price p is feasible for the buyer if and only if $p \leq \bar{p}_b$. When $B_b \geq v_b$ (the common configuration in our experiments), the budget is slack and only the value binds.

The simulator tracks v_b and B_b separately: v_b enters the surplus calculation ($v_b - p$), while B_b enters the hard constraint check. However, the LLM agent’s prompt receives only the collapsed ceiling $\bar{p}_b$ as a single “maximum price” figure; the agent cannot reason about the value–budget distinction. Rule-based agents likewise operate on $\bar{p}_b$ directly. The separation therefore serves the simulator’s enforcement and metrics layers rather than the agent’s decision-making.

Seller. A seller is characterised by a cost c_s (reservation price, representing the minimum acceptable sale price) and a target margin m_s (aspirational markup over cost). An offer at price p is feasible for the seller if and only if $p \geq c_s$. The target margin is used only by the rule-based agent to compute its opening price ($c_s \times (1 + 2m_s)$); it is not enforced as a hard constraint and does not appear in the LLM agent’s prompt. LLM sellers receive only c_s as their “minimum price.”

Zone of possible agreement. The ZOPA [12] for a buyer–seller pair is defined as

$$\text{ZOPA} = v_b - c_s.$$

A deal is structurally possible only when $\text{ZOPA} > 0$. Budget constraints may further tighten the effective ZOPA to $\bar{p}_b - c_s = \min(v_b, B_b) - c_s$. When the

ZOPA is zero or negative, no agreement can satisfy both agents’ hard constraints regardless of strategy.

3.3 Outcome Definitions

Session-level outcomes. When a session terminates with acceptance at price p , the following quantities are computed:

$$\text{buyer surplus} = v_b - p, \tag{1}$$

$$\text{seller surplus} = p - c_s, \tag{2}$$

$$\text{total welfare} = v_b - c_s. \tag{3}$$

Total welfare depends only on the buyer’s value and the seller’s cost, not on the agreed price; the price determines only the distribution of surplus between the two parties. When no deal is reached, all surplus quantities are zero. Each session additionally records the termination reason (ACCEPTED, REJECTED, or TIMEOUT) and the number of rounds taken.

Market-level outcomes. At the market level, outcomes aggregate over sessions within a tick: mean transaction price, within-tick price dispersion (standard deviation), and liquidity (fraction of matched pairs that reach a deal). Two additional metrics support mechanism-comparison experiments: *allocative efficiency*, defined as the ratio of realised total welfare to the theoretical maximum welfare across all matched pairs, and the *surplus Gini coefficient*, measuring inequality in surplus distribution across deals.

3.4 Research Questions

The experiments conducted on the simulator operate at three levels of analysis: micro-level agent bargaining behaviour (Experiments 1, 2), macro-level market dynamics (Experiments 3, 4, 6, and the Experiment 4 anchor-environment extension), and institutional mechanism effects (Experiment 5). Each research question asks whether an observable pattern in the simulator is consistent with a specific prediction from classical theory. Specific hypotheses for each experiment are stated in Section 5.

Micro-level.

- RQ-1.** Do LLM agents exhibit systematic concession patterns across negotiation rounds, and how do concession trajectories differ compared to rule-based agents [14]?
- RQ-2.** Does varying the negotiation horizon affect settlement timing, consistent with deadline-pressure effects observed in human bargaining [13]?

Macro-level.

- RQ-3.** Does a market of LLM bilateral negotiators converge toward a stable price level, and does within-tick price dispersion persist [15]?
- RQ-4.** Do stochastic external shocks produce measurable shifts in market price and liquidity in theoretically predicted directions?
- RQ-6.** Do structural supply-demand shifts (deterministic increases to buyer values or seller costs) produce directionally predicted price and liquidity responses?

Informational environment.

- RQ-4ext.** Does the environmental reference price signal in the agent prompt function as a price anchor that shifts mean transaction prices independently of private fundamentals, and does it modulate the market’s responsiveness to stochastic shocks?

Institutional.

- RQ-5.** Does the choice of buyer–seller matching algorithm affect market-level deal rates, allocative efficiency, and surplus distribution, independently of agent behaviour?

4 Simulator Architecture

The following subsections describe the simulator that implements the setting formalised above: its overall architecture, the agent types used in the reported experiments, the validation pipeline, the market structure, metrics, and reproducibility mechanisms.

4.1 Overall System Architecture

Experiments are configured via YAML files and CLI overrides, which specify agent types, parameter distributions, matching strategies, and seed lists. At runtime, a market simulator orchestrates execution tick by tick: each tick generates buyer and seller populations from the configured distributions, applies optional shocks, delegates pair formation to a pluggable matcher, and dispatches each matched pair to a negotiation session. Each session implements the alternating-offer protocol, invoking agents in turn and routing their outputs through a validation pipeline before updating session state. Results are written to a structured JSONL event log; run-level summaries are computed after all sessions complete. This separation allows agent types, matching strategies, and prompt configurations to be varied independently without modifying the negotiation protocol.

4.2 Agent Types

All agents share a common interface: given a context containing the item under negotiation, the agent’s role and private constraints, the current round, the negotiation history, and the opponent’s last offer, the agent returns a structured action. As noted in Section 3.2, LLM agent prompts present only the effective price ceiling $\bar{p}_b = \min(\text{value}, \text{budget})$ for buyers and cost for sellers. The two agent types used in all reported experiments are described below.

LLM free-language (primary agent). A natural-language agent that generates free-form dialogue rather than structured JSON. The agent receives a prompt containing its role, private constraints, negotiation history, and deadline information; it responds with a free-text message from which the parser extracts a price tag (delimited by `### PRICE($X) ###` markers) and an optional acceptance keyword (`MAKE DEAL`). This design avoids the JSON-parse failures observed with smaller models and is the agent type used in all reported experiments. If the first response fails to yield an extractable price or acceptance signal, the rethink loop (Section 4.3.1) issues one retry with a corrective hint.

Rule-based baseline. A deterministic agent following a linear concession strategy. Buyers open at 50% of their price ceiling and concede linearly toward it over the available rounds; sellers open at $\text{cost} \times (1 + 2m)$, where m is the target margin (default 0.15), and concede toward cost. Each round, the agent computes a concession target as

$$t_r = p_0 + (p_{\max} - p_0) \cdot \frac{r}{R - 1}$$

where p_0 is the opening price, $p_{\max}$ is the constraint boundary, r is the current round, and R is the maximum number of rounds. The agent accepts when the opponent’s standing offer meets or exceeds this target. On the final round, agents accept any offer within their hard constraints to avoid deadlock. This agent produces zero variance across seeds, providing a deterministic baseline for isolating LLM-specific effects. It is used as a comparison condition in Experiment 1 only.

Implementation note. Additional prototype agent architectures were explored during development but are not evaluated in this study.

4.3 Constraint Handling and Feasibility Enforcement

4.3.1 Action Validation Pipeline

Every action produced by an LLM agent passes through a four-stage pipeline before updating session state. Rule-based agents skip Stages 1 and 2, as they produce deterministic, constraint-respecting actions by construction.

Stage 1: Output extraction. The parser extracts a price tag from delimited markers (`### PRICE($X) ###`) and detects an acceptance keyword (`MAKE_DEAL`) in the agent’s prose output, mapping the result to a structured `OFFER`, `COUNTER`, or `ACCEPT` action.

Stage 2: Rethink loop. If extraction fails or the parsed action violates the agent’s own private constraints, one rethink attempt is issued: the original prompt is extended with a corrective message naming the specific error. A successful rethink response is returned as the agent’s action; any remaining constraint violation is left for Stage 3. If both attempts fail to parse, a `REJECT` action is returned, terminating the session.

Stage 3: Protocol and constraint validation. The `ActionJudge` applies domain-level checks. First-round corrections handle protocol violations (e.g. a `COUNTER` on round 0 is relabelled as `OFFER`). Constraint validation checks hard limits: buyer offers must not exceed budget or value; seller offers must not fall below cost; all prices must lie within configurable global bounds $[p_{\min}, p_{\max}]$.

Stage 4: Enforcement. If validation fails, the judge replaces the action with `REJECT` and emits a structured risk event recording the violation type, attempted action, round number, and time step. Risk events are aggregated in the run summary, enabling post-hoc analysis of constraint-violation frequency across conditions.

4.3.2 Feasibility Gate

The simulator includes a configurable deterministic settlement mechanism (`gate_enabled`), which, when enabled, overrides agent decisions by automatically accepting feasible offers. Preliminary development runs showed that this gate drove a substantial fraction of acceptances and suppressed the behavioural dynamics the experiments were designed to study. The gate is therefore *disabled in all reported experiments*, so that every acceptance is LLM-driven. The implications of this design choice are discussed in Section 7.

4.4 Market Layer

Population generation. The simulator supports two scenario modes. In *distribution mode*, buyer values, budgets, seller costs, and target margins are drawn independently from uniform distributions with configurable bounds (e.g. buyer value $\in [\$80, \$150]$, seller cost $\in [\$50, \$120]$ in Experiments 3 and 5; ranges differ across experiments as detailed in Section 5). In *fixed mode*, all agents receive identical parameter values specified in the configuration, enabling controlled single-variable experiments. In both modes, populations are regenerated each tick from a forked RNG, so agents within a tick are independent draws.

Matching strategies. Pair formation is delegated to a pluggable `Matcher` interface. The two matching strategies used in the reported experiments are:

- **Random** (default). Shuffles buyers and sellers independently and pairs them 1:1. Unmatched agents (when population sizes differ) are dropped. This is the baseline matching mechanism used in Experiments 1–4, 6, and the Experiment 4 extension.
- **Surplus-maximising** (oracle). Computes $ZOPA = \text{buyer value} - \text{seller cost}$ for all possible pairs, then greedily assigns pairs in descending ZOPA order (each agent used at most once). Only pairs with positive ZOPA are matched. This uses private information and represents a theoretical upper bound on matching efficiency, not a realistic mechanism. Used as the treatment condition in Experiment 5.

Demand and supply shocks. An optional shock mechanism applies multiplicative perturbations to buyer values and seller costs at the beginning of each tick, with a 30% trigger probability (as configured in Experiment 4). When activated, buyer values are scaled by a demand multiplier drawn uniformly from a specified range (e.g., $[0.7, 1.3]$), simulating shifts in item desirability. Seller costs are independently scaled by a supply multiplier from the same range, while target margins remain constant.

Buyer budgets are exempt from these shocks; the budget serves as an inelastic spending ceiling. Consequently, positive demand shocks are partially muted for budget-constrained buyers whose shocked value exceeds their available funds. This deliberate asymmetry preserves the role of the budget as a hard constraint while allowing the agent’s willingness-to-pay to fluctuate with market conditions.

4.5 Metrics and Logging

Session-level metrics. Each completed negotiation produces a structured result containing: deal status, agreed price (if any), number of rounds, termination reason, buyer and seller surplus, and a list of risk events. Run-level aggregates include deal success rate, price statistics (mean, median, standard deviation), mean surplus by role, deadlock rate, and constraint-violation counts.

Tick-level and efficiency metrics (market mode). In market mode, per-tick aggregates—including mean transaction price, within-tick price standard deviation, and liquidity—are computed after each tick’s sessions complete and logged as `tick.end` events, enabling time-series analysis. Two additional metrics support mechanism-comparison experiments: *allocative efficiency* (ratio of realised welfare to theoretical maximum across all matched pairs) and the *surplus Gini coefficient* (inequality in surplus distribution across deals).

Event logging. Every negotiation action, session outcome, constraint violation, and tick-level aggregate is written to a newline-delimited JSON log (`events.jsonl`). For LLM agents, each turn event additionally records the intended action, the effective action, an override flag, and the feasibility threshold. This granular logging supports detailed post-hoc analysis without re-execution.

4.6 Reproducibility

All stochastic market-level processes draw from a `SeededRNG` wrapper that forks a child seed at each tick, ensuring full reproducibility of population generation, matching, and shock application given the same seed. LLM inference introduces irreducible non-determinism from the model itself. All experimental parameters are specified in YAML files, and each experiment runner records the active configuration, the git commit hash, and the seed list in a structured JSON summary.

5 Methodology and Experimental Design

This section describes the methodology and experimental design for all experiments conducted on the platform. The experiments are organised into three levels of analysis—micro-level agent bargaining behaviour (Experiments 1, 2), macro-level market outcomes (Experiments 3, 4, 6, and the Experiment 4 anchor-environment extension), and institutional mechanism effects (Experiment 5)—reflecting the platform’s ability to connect individual negotiation dynamics to aggregate market properties within a single system. Results are reported in Section 6.

5.1 Experimental Philosophy

The experimental programme is designed around four principles.

First, *controlled comparison against theory*. Each experiment targets a specific prediction from classical bargaining or market theory and varies a single independent variable while holding all other conditions constant. This enables directional comparison against theoretical baselines.

Second, *micro-to-macro connection*. The platform supports experiments at both the individual session level (where agent behaviour can be observed turn-by-turn) and the market level (where aggregate statistics emerge from many sessions per tick). This multi-level design enables investigation of whether micro-level bargaining patterns translate into macro-level market regularities.

Third, *transparent protocol reporting*. The bargaining protocol and prompt architecture are active throughout all experiments and shape the action space available to agents. Rather than treating these as invisible background conditions, we report their design rationale and discuss their implications for interpretation. Prompt architecture is treated as a fixed protocol parameter in the current study; ablation is identified as a priority for future work.

Fourth, *behavioural interpretability*. Wherever possible, we connect observed outcomes—deal success rates, concession speeds, surplus splits—to behavioural patterns documented in the human negotiation literature, asking whether LLM agents produce qualitatively similar dynamics under equivalent structural conditions. This framing enables interpretation of results through both an engineering lens (does the simulator work?) and a behavioural lens (do the agents negotiate in recognisable ways?).

5.2 Common Setup

All experiments share the following configuration unless explicitly overridden.

Model and inference. All experiments use `Qwen/Qwen2.5-14B-Instruct` served via HuggingFace Transformers at temperature 0.2, with a maximum output length of 512 tokens and a 60-second timeout per call. Failed LLM calls are retried once with a format-correction prompt (the rethink loop described in Section 4.3.1). If the retry also fails to parse, a `REJECT` action is returned, terminating the session without a deal; if the retry parses but remains infeasible, the action is passed through to the `ActionJudge` for enforcement.

Agent type. All LLM conditions use the `llm_free_language` agent type, which generates natural-language prose and extracts price tags post-hoc (Section 4.2). The rule-based baseline agent is used as a comparison condition in Experiment 1 only.

Feasibility gate. The deterministic settlement gate (Section 4.3.2) is *disabled* in all experiments, so that acceptances are LLM-driven rather than gate-overridden. Preliminary development runs with the gate enabled showed that it drove a substantial fraction of acceptances (estimated at roughly 40% in early pilot trials) and suppressed behavioural signals such as deadline effects and extended concession trajectories, motivating the gate-off design adopted throughout.

Prompt design All experiments reported in this thesis use the *free-language* prompt pipeline, in which agents respond in natural language rather than structured JSON. Each prompt is assembled from a fixed sequence of components, described below in the order they appear. The buyer and seller prompts are symmetric; we describe the buyer version and note differences where they arise.

1. **Role assignment.** The prompt opens by assigning the agent a role and persona: “*You are Buyer, a buyer looking for a good deal on <item>. You are polite, strategic, and want to get the best price within your budget.*” The seller variant instead instructs the agent to “*maximize profit ... while being reasonable.*”

2. **Product information.** The item name and, when available, the market reference price are stated (e.g. “*Product: Widget (market reference price: \$100.00)*”). The reference price serves as a shared anchor visible to both parties.
3. **Round state.** A single line states the current round number, the total round limit, and the number of remaining rounds (e.g. “*Round 3 of 10 (7 remaining)*”).
4. **Conversation history.** All prior turns are rendered as a natural-language transcript, with each entry formatted as “[*Round n*] *BUYER/SELLER: <message>*”. This gives the model full visibility of the negotiation dialogue so far.

5. **Decision hint.** A dynamic block computes the utility of the opponent’s standing offer and presents the result explicitly. When the offer is individually rational, the hint reads:

“The seller’s last price is \$X. Your maximum is \$Y. This price is WITHIN your limit and you would gain \$Z surplus. Consider accepting with MAKE DEAL if it is close to the price you would counter with.”

When the offer exceeds the agent’s limit, the hint instead states:

“This price EXCEEDS your limit—you must counter below \$Y.”

6. **Monotonic concession guidance.** The prompt guides agents toward monotonic concession: buyers are encouraged to raise each successive offer, and sellers to lower theirs. This design reflects the standard assumption in alternating-offer bargaining, where rational agents do not retract concessions already made [14]. The prompt conveys this through directional suggestions such as “*Each counter-offer you make should be HIGHER than your previous offer (concede upward toward agreement),*” and advises against crossing the opponent’s last price. Agents are also encouraged to accept rather than make marginal counteroffers when the opponent’s price is already close to what they would propose.

In practice, LLM agents follow this guidance consistently, so the *direction* of the concession trajectory is largely a product of prompt design rather than emergent behaviour. The concession *magnitude* and *timing*—such as Boulware-style early conservatism or deadline-driven acceleration—remain attributable to the model, since the guidance constrains direction but not step size or pace.

7. **Private constraints.** The agent’s reservation price is stated as confidential:

“Your maximum acceptable price is \$Y (confidential, do not reveal).”

For sellers, the prompt additionally provides a computed initial asking price derived from the reference price and target margin, giving the model a concrete starting point.

8. **Deadline salience (configurable).** When enabled, escalating urgency cues are injected as the round limit approaches: a mild reminder when one-third of rounds remain (“*Consider conceding to reach a deal*”), a warning at two rounds remaining (“*Only N rounds left. No deal = zero surplus*”), and a final-round ultimatum (“*FINAL ROUND. Accept any offer within your limits or get NOTHING.*”). This component is the manipulated variable in Experiment 2.
9. **Price-tag format.** The prompt requires exactly one price offer per turn using an extractable tag: `### BUYER_PRICE($X) ###` (or `SELLER_PRICE` for sellers). Two worked examples demonstrate correct usage. A downstream regex parser extracts the price from this tag, decoupling price extraction from free-text generation.
10. **Deal acceptance protocol.** A separate block defines the semantics of `MAKE_DEAL`: it means accepting the opponent’s last price exactly as offered, without including a new price tag. The prompt explicitly distinguishes acceptance from counter-offering with worked examples for each case, reducing ambiguity for the model.
11. **Communication strategy (configurable).** An optional preamble—*assertive*, *collaborative*, or *strategic*—instructs the model on tone and messaging tactics. For example, the strategic condition tells the agent to “*exaggerate how close you are to your limit*” and “*use the message to influence the opponent’s expectations.*”

The prompt closes with a generation cue (“*Now, respond as Buyer:*”) that triggers the model’s reply. The full prompt is assembled fresh on every turn with updated round state, history, and decision hint, so the model always receives current information without relying on its own memory across turns.

Reproducibility. All stochastic processes are controlled by a `SeededRNG` with per-tick forking (Section 4.6). Each experiment records the active configuration, seed list, and git commit hash in a structured JSON summary. Every agent action, session outcome, and constraint violation is logged to a JSONL event stream.

Replication. All experiments are replicated across three random seeds (42, 123, 456). Table 1 summarises all experiments.

5.3 Micro-Level Experiments: Agent Bargaining Behaviour

Experiments 1 and 2 use fixed-scenario parameters to isolate individual agent behaviour from market-level variation. Each experiment tests whether LLM agents exhibit behaviour directionally consistent with a specific prediction from classical bargaining theory: systematic concession dynamics [14] and deadline-driven concession acceleration [13].

Table 1: Summary of all experiments, organised by level of analysis.

Level	Exp	Name	Independent variable	Agent type	Sessions	Seeds
Micro	1	Concession	agent type (2 levels)	Rule-based / LLM	144	3
	2	Deadline	max rounds (3 levels)	LLM	216	3
Macro	3	Market	time (10 ticks)	LLM	300	3
	4	Shock (stochastic)	shock on/off	LLM	600	3
	4ext	Anchor environment	anchor mode (3 levels) $\times$ shock	LLM	1,200	3
	6	Supply-demand shift	supply/demand shift (3 levels)	LLM	576	3
Institutional	5	Mechanism	matching algorithm (2 levels)	LLM	384	3
Total					3,420	

5.3.1 Experiment 1: Concession Dynamics

Theoretical motivation. Classical bargaining theory identifies two archetypal concession strategies: the Boulware strategy, in which an negotiator opens aggressively and concedes minimally until the final rounds (producing a concave offer trajectory), and the Conceder strategy, in which an negotiator makes large early concessions and approaches the settlement zone quickly (producing a convex trajectory) [12]. In Rubinstein’s sequential bargaining model [14], more patient agents—those who discount the future at a lower rate—extract a larger share of the surplus, because their willingness to wait signals that delay is less costly to them than to the opponent. LLM agents, however, possess neither explicit utility functions nor discount rates; they receive cost and value information through their prompts and generate offers through next-token prediction. The central question is therefore whether LLM agents produce recognisable concession patterns, or whether their offer trajectories are essentially arbitrary within the feasible range. The rule-based agent in this experiment serves as a *behavioural baseline*: it follows a known, deterministic linear concession schedule, so any systematic departure in the LLM agent’s trajectory can be attributed to the LLM’s decision-making process, subject to the prompt instruction described in Section 5.2.

Research questions.

- RQ-11.** Do LLM agents exhibit systematic concession patterns across negotiation rounds?
- RQ-12.** How do concession trajectories differ between rule-based and `llm_free_language` agents?
- RQ-13.** Does the `llm_free_language` agent produce a distinctive concession trajectory compared to the deterministic rule-based agent?

Independent variable. Agent type, varied across two conditions: `rule_based` and `llm_free_language`. Both buyer and seller use the same agent type within each condition.

Controlled variables. All sessions use fixed-mode parameters to isolate the effect of agent type from scenario variation. The fixed scenario is: buyer value = \$120, seller cost = \$80, buyer budget = \$130, item reference price = \$100, seller target margin = 15%. This creates a ZOPA of \$40 (\$120 − \$80), wide enough for reliable settlement. Maximum rounds = 10. Feasibility gate is disabled.

Design. 2 agent-type conditions × 3 seeds × 24 sessions per seed = 144 sessions total.

Metrics.

- Deal rate (%) per condition.
- Mean agreed price (\$ ± standard deviation across seeds).
- Mean rounds to settlement.
- Per-round offer price trajectory (concession curve), plotted per agent type and role.
- Buyer and seller surplus.

Hypotheses.

- H-1a.** Both agent types achieve >90% deal rate given the wide ZOPA.
- H-1b.** LLM agents settle in fewer rounds than the rule-based agent, whose linear concession formula spreads concessions across the full horizon.
- H-1c.** The `llm_free_language` agent produces a concave (Boulware-style) concession trajectory, in contrast to the linear trajectory of the rule-based agent.

5.3.2 Experiment 2: Deadline Pressure

Theoretical motivation. Roth et al. [13] documented that human bargainers make large concessions near deadlines, with many agreements clustering in the final moments of the available horizon. As the deadline approaches, the probability of reaching no agreement increases, then human negotiators have higher tendency to close deal. LLM agents lack this kind of subjective risk aversion since they do not experience disutility from delay. However, they do receive explicit deadline information through the prompt’s deadline-salience language, which warns them when few rounds remain. This experiment therefore tests a

narrower question: whether the *structural availability* of deadline information is sufficient to produce deadline-consistent behaviour, even in the absence of the psychological mechanisms that drive the effect in human subjects.

Research questions.

RQ-21. Does varying the negotiation horizon (maximum rounds) affect when deals settle?

RQ-22. Do LLM agents exhibit deadline effects—accelerated concession near the final rounds—consistent with the findings of Roth et al. [13]?

Independent variable. Maximum rounds per session, varied across three levels: 6, 12, and 20.

Controlled variables. Agent type is `llm_free_language` in all conditions. Fixed scenario parameters match Experiment 1 (buyer value = \$120, seller cost = \$80, ZOPA = \$40). Feasibility gate is disabled. The prompt includes configurable deadline-salience language that warns the agent when few rounds remain.

Design. 3 horizon conditions $\times$ 3 seeds $\times$ 24 sessions per seed = 216 sessions total.

Metrics.

- Deal rate per horizon condition.
- Mean settlement round per condition (mean $\pm$ standard deviation).
- Settlement round as a fraction of maximum rounds (normalised timing).
- Proportion of agreements occurring in the last two rounds of the available horizon.

Hypotheses.

H-2a. If agents are sensitive to deadlines, longer negotiation horizons should lead to later settlements. Agreements should also become more concentrated near the deadline itself.

5.4 Macro-Level Experiments: Market Outcomes

Experiments 3, 4, 6, and the Experiment 4 anchor-environment extension shift the unit of analysis from individual sessions to multi-tick markets populated by LLM agents. These experiments test whether aggregate market outcomes are consistent with predictions from bilateral bargaining theory [15] and supply-demand theory. All use distribution-mode parameters, random matching, and `llm_free_language` agents.

5.4.1 Experiment 3: Market Price Discovery

Theoretical motivation. Rubinstein and Wolinsky [15] show that price convergence in decentralised bilateral markets requires specific structural conditions: sufficiently patient agents and sufficient market thickness. When these conditions are absent, persistent price dispersion across independently negotiated pairs is expected in theory. The present simulator operates as a collection of independent bilateral sessions with private constraints. Within each tick, agents do not observe other pairs’ offers or outcomes and there is no centralised price signal. This design means that the Law of One Price should *not* hold, and any price convergence across ticks would reflect emergent regularities in how LLM agents use their private information rather than any equilibrating market mechanism. Experiment 3 tests whether LLM-populated markets exhibit these structural properties: stable mean prices along with persistent within-tick dispersion, and whether any convergence pressure emerges over the ten-tick observation window.

Research questions.

- RQ-31.** Does a multi-tick market populated by LLM bilateral negotiators converge toward a stable price level?
- RQ-32.** Does price dispersion persist within ticks, consistent with the bilateral bargaining prediction that the Law of One Price fails when pairs negotiate independently [15]?

Independent variable. Time, observed longitudinally over 10 market ticks. There is no between-condition manipulation; this is a single-condition observational study replicated across three seeds.

Controlled variables. The simulator runs in market mode with 10 buyer–seller pairs per tick, using random matching. Agent type is `llm_free_language`. Scenario mode is *distribution*: buyer values are drawn uniformly from [\$80, \$150], seller costs from [\$50, \$120], buyer budgets from [\$100, \$200], and seller margins from [5%, 25%]. Maximum rounds = 10. Feasibility gate is disabled.

Design. 3 seeds (42, 123, 456) $\times$ 10 ticks $\times$ 10 sessions per tick = 300 sessions total.

Metrics.

- Mean transaction price per tick, with within-tick standard deviation.
- Liquidity per tick: fraction of matched pairs that reach a deal.
- Price trend: slope of a simple linear regression of mean price over ticks.
- Mean buyer surplus and mean seller surplus per tick.

- Early-vs-late comparison: mean price and liquidity in ticks 0–2 versus ticks 7–9.

Hypotheses.

- H-3a.** Mean price stabilises within the observed horizon, with a near-zero price trend slope across ticks.
- H-3b.** Within-tick price dispersion remains substantial (standard deviation $> \$10$), consistent with bilateral bargaining theory.
- H-3c.** Buyer surplus exceeds seller surplus on average, reflecting the buyer’s first-mover advantage in the alternating-offer protocol.

5.4.2 Experiment 4: Stochastic Shock Response

Theoretical motivation. In competitive market models, prices adjust endogenously to clear supply-demand imbalances: demand increases push prices upward, supply increases push them downward, and the adjustment occurs through the incentives of self-interested participants. In the present simulator, no such endogenous adjustment mechanism exists at the market level; agents negotiate bilaterally on the basis of their private constraints and are unaware of market-wide conditions. Experiment 4 therefore tests whether LLM agents exhibit any *emergent* price adjustment in response to exogenous changes in fundamentals, or whether price-level effects are entirely mechanical consequences of ZOPA widening or compression.

Research questions.

- RQ-41.** Do exogenous demand and supply shocks produce measurable shifts in market price and liquidity?
- RQ-42.** Is the direction of price adjustment consistent with economic theory (demand increase $\rightarrow$ price increase; supply increase $\rightarrow$ price decrease)?

Independent variable. Shock condition, varied across two levels: `no_shock` (baseline) and `with_shock`. In the shock condition, each tick has a 30% probability of a shock event. When a shock fires, buyer values are scaled by a demand multiplier drawn uniformly from $[0.7, 1.3]$ and seller costs are scaled by an independently drawn supply multiplier from the same range. Budgets are not modified.

Controlled variables. Both conditions use the same tick structure as Experiment 3 (10 ticks, 10 pairs per tick, random matching, `llm_free_language` agents, gate disabled), but with different base parameter ranges (buyer values from $[\$90, \$170]$, seller costs from $[\$40, \$90]$) and `coherent_sampling` disabled to preserve the shock signal.

Design. 2 conditions $\times$ 3 seeds (42, 123, 456) $\times$ 10 ticks $\times$ 10 sessions per tick = 600 sessions total (300 per condition).

Metrics.

- Mean transaction price per tick, compared across conditions.
- Within-tick price standard deviation per condition.
- Mean liquidity per condition, with minimum and maximum tick-level values.
- Price and liquidity differences (Δ) between shock and no-shock conditions.

Hypotheses.

- H-4a.** The shock condition produces higher price volatility (larger tick-to-tick price variance) than the baseline.
- H-4b.** Mean price and mean liquidity are lower under shocks, because negative shocks (demand multiplier < 1 or supply multiplier > 1) compress the ZOPA for affected pairs.
- H-4c.** Worst-case liquidity drops more severely under shocks than best-case liquidity improves, producing an asymmetric tail risk.

From shock response to anchor environment. Experiment 4 was designed with the item reference price treated as a fixed background constant. During analysis, this constant emerged as a candidate confound: the price anchor that agents receive in the prompt may itself shape the price level and the magnitude of shock effects, independently of the underlying value and cost distributions. The anchor-environment extension (Section 5.4.3) addresses this by promoting the reference price to an explicit independent variable while retaining the same market configuration. Because the fixed-anchor condition in the extension is identical in setup to Experiment 4, the Experiment 4 dataset is reused as the fixed-anchor arm without re-running sessions.

5.4.3 Experiment 4 Extension: Anchor Environment

The item reference price included in the agent prompt may function as an environmental price signal that shapes both price levels and shock responsiveness. This extension is designed to probe the *mechanism* behind this potential price stickiness: if agents anchor to the prompt-embedded reference price rather than discovering prices from their private fundamentals alone, then varying the reference signal should produce large price-level effects even when buyer value and seller cost distributions are held identical across conditions. Such an outcome would identify prompt-level information as a first-order determinant of market outcomes, with direct implications for how LLM-based market simulations

should be designed and interpreted—in particular, whether a fixed reference price inadvertently imposes a nominal anchor on all sessions. Crossing anchor mode with the shock condition from Experiment 4 additionally enables a *moderation analysis*: if the informational environment moderates the market’s responsiveness to fundamental shocks, the magnitude of the shock price effect should differ systematically across anchor conditions. This extension tests the hypothesis by varying the anchor mode across three conditions while crossing each with the shock/no-shock manipulation from Experiment 4, using the same market parameters and seed schedule.

Research questions.

- RQ-4ext1.** Does the environmental reference price in the agent prompt function as a price anchor that shifts mean transaction prices independently of private fundamentals?
- RQ-4ext2.** Does anchor mode modulate the market’s responsiveness to stochastic demand and supply shocks?

Independent variable. Anchor mode, varied across three levels:

- **fixed_anchor**: reference price held constant throughout all ticks (the default in Experiment 4).
- **updated_anchor**: reference price dynamically updated after each tick to the previous tick’s mean deal price.
- **no_anchor**: reference price omitted from the prompt entirely.

Each anchor mode is crossed with the shock condition (shock on/off), yielding six experimental cells.

Controlled variables. All conditions use the same market parameters as Experiment 4: 10 ticks, 10 pairs per tick, `llm_free_language` agents, gate disabled, distribution-mode parameters with `coherent_sampling` off.

Design. 3 anchor modes $\times$ 2 shock conditions $\times$ 3 seeds $\times$ 10 ticks $\times$ 10 pairs per tick = 1,800 sessions in total across the three modes, of which the **fixed_anchor** condition reuses the Experiment 4 dataset (600 sessions), as established above, and the **updated_anchor** and **no_anchor** conditions contribute 1,200 new sessions.

Hypotheses.

- H-4ext-a.** Different anchor modes will produce substantially different mean transaction prices despite identical buyer value and seller cost distributions, indicating that the reference signal shapes price levels above and beyond private constraints.

H-4ext-b. The `no_anchor` condition will exhibit lower shock sensitivity than anchored conditions, since agents without a reference signal will have less informational material to propagate shock-induced price changes.

Metrics.

- Mean transaction price per anchor mode and shock condition.
- Within-condition price standard deviation.
- Market liquidity per condition.
- Shock price effect: difference in mean deal price between with-shock and no-shock conditions within each anchor mode.

5.4.4 Experiment 6: Structural Supply-Demand Shifts

Experiment 4 applies stochastic multiplicative shocks of unknown direction. Experiment 6 complements this with deterministic, directional shifts to buyer values or seller costs, enabling cleaner tests of whether LLM-agent markets respond to supply and demand conditions in theoretically predicted directions.

Theoretical motivation. Standard supply-demand theory predicts an asymmetry between demand shocks and supply shocks in their effect on transaction prices. A seller cost increase (supply shock) constitutes a hard constraint: no deal can occur below cost, so higher costs must translate into higher prices or into deal failure. Price pass-through is therefore structurally compelled. A buyer value increase (demand shock), by contrast, creates room for deals at higher prices but does not compel prices upward—buyers retain the option to accept at their previous price levels if sellers do not demand more. This asymmetry implies that supply shocks should produce *stronger* price pass-through than demand shocks of equal magnitude. Experiment 6 tests whether LLM agents reproduce this theoretical asymmetry, or whether they exhibit a different pattern—for instance, because buyers respond to their increased private value by bidding more aggressively, generating demand-side price pressure that partially offsets the asymmetry.

Research questions.

- RQ-61.** Do transaction prices shift in the direction predicted by supply-demand theory when buyer values increase (demand shift) or seller costs increase (supply shift)?
- RQ-62.** Does the magnitude of price and liquidity response differ between demand-side and supply-side shifts?
- RQ-63.** Can observed liquidity changes be decomposed into mechanical effects (fewer feasible pairs) and behavioural effects (lower deal success among feasible pairs)?

Independent variable. Supply-demand condition, varied across three levels:

- **baseline:** buyer values drawn from [\$90, \$170], seller costs from [\$40, \$90].
- **demand_shock:** buyer values shifted upward by \$20, drawn from [\$110, \$190]; seller costs unchanged.
- **supply_shock:** seller costs shifted upward by \$20, drawn from [\$60, \$110]; buyer values unchanged.

Controlled variables. The simulator runs in market mode with 8 buyer-seller pairs per tick over 8 ticks. Agent type is `llm_free_language`. Random matching. Maximum rounds = 10. Feasibility gate is disabled.

Design. 3 conditions $\times$ 3 seeds (42, 123, 456) $\times$ 8 ticks $\times$ 8 pairs per tick = 576 sessions total (192 per condition).

Metrics.

- Mean transaction price per condition.
- Liquidity (deal rate) per condition.
- Total welfare, mean buyer surplus, and mean seller surplus per condition.
- Within-condition price dispersion (standard deviation).
- Feasibility decomposition: fraction of pairs with positive ZOPA (mechanical), deal success rate among feasible pairs (behavioural), and overall liquidity.

Hypotheses.

- H-6a.** A demand shift (higher buyer values) increases mean transaction price and liquidity: more pairs have positive ZOPA, and buyers with higher values accept higher prices.
- H-6b.** A supply shift (higher seller costs) increases mean transaction price but decreases liquidity: sellers demand higher prices to cover costs, and fewer pairs have sufficient ZOPA for agreement.
- H-6c.** Liquidity effects are partly mechanical (ZOPA compression) and partly behavioural (LLM agents are less effective at reaching agreement when the ZOPA is narrow).

5.5 Institutional Experiment: Mechanism Design

Experiment 5 tests whether an institutional design choice—the buyer–seller matching algorithm—affects market-level outcomes independently of agent behaviour. This complements the micro-level and macro-level experiments by examining whether market structure can be improved through mechanism design even when agent sophistication is held constant.

In the mechanism design literature, the matching rule is understood as an *institutional rule* that determines which agents interact, independently of how those agents behave once matched. It is well established in market design that the choice of matching mechanism can substantially affect allocative efficiency, even when all participants are equally strategic. Experiment 5 tests whether this classical insight—that institutional rules shape aggregate outcomes beyond participant behaviour—extends to a setting where participants are LLM agents rather than human or game-theoretically rational actors. If the surplus-maximising matcher produces higher deal rates and efficiency than random matching *without* any change to agent behaviour, this provides direct evidence that institutional design is a lever for improving LLM-populated market performance that operates independently of prompt engineering or model capability.

5.5.1 Experiment 5: Matching Mechanism Comparison

Research question. Does the matching algorithm affect market-level deal rates, allocative efficiency, welfare, and price dispersion?

Independent variable. Matching algorithm, varied across two conditions: **random** (baseline, all agents paired regardless of ZOPA) and **surplus_max** (oracle, greedy maximum-ZOPA pairing that only matches pairs with positive ZOPA overlap). Additional matchers (sorted heuristic, round-robin) are implemented but not evaluated in this study.

Controlled variables. The simulator runs in market mode with 8 buyer–seller pairs per tick over 8 ticks. Agent type is `llm_free_language`. Scenario mode is *distribution*, using the same parameter ranges as Experiment 3 (buyer values from [\$80, \$150], seller costs from [\$50, \$120]). Maximum rounds = 10. Feasibility gate is disabled.

Design. 2 matching conditions $\times$ 3 seeds (42, 123, 456) $\times$ 8 ticks $\times$ 8 pairs per tick = 384 nominal sessions (335 after surplus-max filtering excludes negative-ZOPA pairs).

Metrics.

- Deal rate per condition.
- Mean transaction price and price standard deviation per condition.

- Allocative efficiency: realised welfare as a fraction of theoretically maximum welfare.
- Surplus Gini coefficient: inequality of surplus distribution across deals.
- Mean buyer surplus and mean seller surplus per condition.

Hypotheses.

- H-5a.** Surplus-maximising matching produces a higher deal rate than random matching, because it eliminates structurally infeasible pairs.
- H-5b.** Surplus-maximising matching produces higher allocative efficiency, because it concentrates trades in high-surplus pairs.

5.6 Deferred Experiments

Three additional experiment families are designed and partially implemented but deferred to future work. An anchoring-effect experiment was intended to vary buyer valuations to test first-offer anchoring [2], but was deferred due to a ZOPA confound that requires redesign: changing buyer values simultaneously shifts the anchor and the feasible range, making effects inseparable. A reputation and repeated-interaction experiment uses the round-robin matcher and reputation agent to test whether access to opponent interaction history improves deal rates over repeated encounters. A communication-strategy experiment uses prompt-level communication modifiers—neutral, assertive, collaborative, and strategic—to test whether message framing affects concession speed and settlement price. Both require the feasibility gate to be disabled so that strategic adaptation can manifest; their evaluation is deferred to future work (Section 8).

5.7 Protocol Parameters and Design Choices

Several protocol-level design choices are held constant across the experimental programme. We enumerate them here so that their effects can be tracked when interpreting results in Section 6.

Prompt architecture. Two prompt-level choices are active in all LLM conditions: the counteroffer protocol (agents counter-offer or accept; episodes that reach the round limit end as no-deal) and the accept-condition block (which pre-computes utility and may reduce anchoring effects). Neither has been individually ablated; their separate effects are unknown.

Single model family. All experiments use `Qwen/Qwen2.5-14B-Instruct` at temperature 0.2. Results may not generalise to other model families, scales, or temperatures; no cross-model comparison has been conducted.

Seed limitations. All experiments use three seeds (42, 123, 456), supporting directional replication but not formal statistical testing. No experiment includes sufficient replications for confidence-interval estimation or hypothesis testing; all findings are reported as descriptive characterisations.

5.8 Evaluation Protocol

For each experiment, we report the following:

1. **Primary outcome metrics** as specified in the experiment design (deal rate, settlement price, surplus, timing, or market-level aggregates).
2. **Deal success rate** as a standard metric across all experiments, enabling systematic comparison of agreement rates across conditions and experimental settings.
3. **Hypothesis evaluation:** whether each stated hypothesis is supported, rejected, or indeterminate, using descriptive comparison (means, standard deviations, Pearson correlation) rather than formal statistical tests.
4. **Confound acknowledgement:** for each finding, whether the result is attributable to agent behaviour, protocol structure, or some combination that the current design cannot separate.

All analyses are descriptive. No hypothesis tests, confidence intervals, or effect sizes are computed. Findings are characterised as “consistent with,” “directionally aligned with,” or “inconsistent with” theoretical predictions, rather than as confirmations or rejections in the statistical sense.

6 Results

This section reports results for Experiments 1–6 (including the Experiment 4 anchor-environment extension) using `Qwen/Qwen2.5-14B-Instruct` with the free-language agent type and the feasibility gate disabled throughout. Section 6.1 validates the experimental backend before any behavioural claims are made. Section 6.2 reports micro-level findings for individual bargaining behaviour. Section 6.3 reports macro-level findings for market outcomes. Section 6.4 synthesises cross-experiment patterns, deal success rates, and behavioural comparisons with an explicit evidence-strength assessment.

6.1 Pipeline Validity and Sanity Checks

Backend validity. The `Qwen2.5-14B-Instruct` model produced valid, parseable responses across all sessions (parse success rate: 100%). The feasibility gate is disabled throughout, so every acceptance is LLM-driven rather than gate-overridden. Across the core six experiments (1–6; 2,220 nominal sessions, 2,171 after surplus-max matcher exclusions), the overall deal rate is 78.6% (1,705

deals); including the Experiment 4 anchor-environment extension (1,200 additional sessions), the total programme comprises 3,420 sessions. Non-deal outcomes are either timeouts (maximum rounds reached without agreement) or implicit rejections through counter-offer exhaustion.

Because the gate is disabled throughout, results are interpretable as agent behaviour rather than as the combined output of agent decisions and deterministic enforcement. The counteroffer protocol, acceptance hints, and monotonic concession instructions remain active as fixed protocol parameters across all conditions (Section 5.7).

6.2 Micro-Level Results: Agent Bargaining Behaviour

Experiments 1 and 2 use fixed-scenario parameters to isolate individual agent behaviour from market-level variation.

6.2.1 Experiment 1: Concession Dynamics

Table 2 reports the full concession trajectory for the rule-based and `llm.free.language` conditions. The scenario is fixed at buyer value = \$120, seller cost = \$80, giving a ZOPA of \$40. All 140 deals across 144 sessions are LLM-driven acceptances; zero gate-driven settlements occur. The deal rate is 97.2% (140/144).

Table 2: Concession trajectories by condition and round ($n = 72$ per condition, 3 seeds). LLM values are mean $\pm$ standard deviation; rule-based values are deterministic (zero variance). Buyer value = \$120, seller cost = \$80.

Condition	Buyer offers (\$)			Seller counters (\$)		
	Round 0	Round 2	Round 4	Round 1	Round 3	Round 5
<code>rule_based</code>	60.00	73.33	86.67	101.33	96.00	90.67
<code>llm.free.language</code>	71.88 ± 3.79	85.18 ± 0.87	86.72 ± 1.93	89.93 ± 0.59	87.60 ± 2.05	88.06 ± 1.87

Finding 1: LLM buyer concession follows a concave trajectory. The LLM free-language buyer opens at \$71.88 (round 0), jumps by \$13.31 to \$85.18 in round 2, then slows sharply: round 4 moves by only \$1.54, with subsequent steps of \$0.31 and \$0.14. This concave deceleration—large initial concession followed by rapid hardening—is consistent with Boulware-style bargaining behaviour, where an agent makes early accommodations but becomes increasingly resistant as the negotiation progresses. The pattern is theoretically coherent: a large initial concession signals genuine willingness to negotiate, while subsequent hardening communicates commitment to a position—a “tough but not foolish” strategy that preserves credibility. The magnitude of the initial jump is notable: at \$13.31, it represents approximately one-third of the full ZOPA width (\$40) in a single move, suggesting that the agent front-loads its flexibility early rather than distributing it evenly. The rule-based buyer, by contrast, concedes at a

fixed linear rate of approximately \$13.33 per round. H-1b is partially supported: LLM agents settle earlier than the rule-based agent, though not because of a more even concession schedule.

Finding 2: Concession burden is asymmetrically borne by the buyer.

Mean deal price is \$87.82. Against the ZOPA midpoint of \$100 (equidistant between buyer value \$120 and seller cost \$80), the buyer captures \$32.18 in surplus (80.5% of ZOPA) while the seller captures only \$7.82 (19.5%). The buyer-to-seller concession ratio is 5.65:1. All deals cluster in the \$80–\$91.25 price range, occupying only the bottom 28% of the available ZOPA. This asymmetry reflects differential concession willingness: the LLM seller exhibits strong anchor rigidity, treating its early counter-offer as a defensible position and resisting further movement, while the buyer bears the concession burden to close the deal. This pattern is behaviourally consistent with the anchoring literature [2], in which the party that establishes an early positional anchor captures a disproportionate share of the surplus. A subtle but important feature of this result is that the effective anchor is the *seller’s* counter-offer rather than the buyer’s opening bid, even though the buyer moves first. The buyer’s initial offer (\$71.88) serves as a starting point for movement, but the seller’s counter at \$89.93 appears to function as the positional anchor around which the final deal is reached—the seller barely moves from it (\$2 total concession) while the buyer closes nearly the entire remaining gap. This is consistent with Galinsky and Mussweiler’s [2] finding that the party whose reference point governs the negotiation captures more surplus, independent of who makes the first move: what matters is whose offer becomes the de facto anchor, not the turn order.

Finding 3: Seller anchors high and concedes slowly. The LLM seller opens at \$89.93 in round 1—notably lower than the rule-based seller’s \$101.33—but then barely moves: rounds 3 and 5 are \$87.60 and \$88.06, a total seller concession of under \$2. Seller variance is also low ($\pm \$0.59$ at round 1), indicating consistent behaviour across seeds. The seller appears to find an early defensible position and resist further movement.

Rule-based baseline observation. The rule-based buyer opens at exactly 50% of its value ceiling (\$60.00) and moves linearly, while the LLM buyer opens \$11.88 higher. The LLM buyer’s willingness to make a more generous opening offer is directionally consistent with H-1c (structured reasoning affects offer behaviour), though a direct comparison between rule-based and free-language conditions is confounded by the complete difference in decision architecture.

6.2.2 Experiment 2: Deadline Pressure

Table 3 reports settlement timing and price outcomes for three time-horizon conditions (maximum rounds: 6, 12, 20) with the gate disabled and `llm_free_language` agents. 216 sessions were run (72 per condition); deal rate is 96.3% (208 deals). All 208 deals are LLM-driven.

Table 3: Deadline experiment outcomes by time horizon ($n = 72$ per condition, 3 seeds). Gate disabled; buyer value = \$120, seller cost = \$80. Concession rate is mean \$/round to settlement.

Max rounds	Deal rate	Mean price (\$)	Avg rounds	Last-2-round share	Concession rate (\$/round)
6	94.4%	86.47	5.21	82.4%	2.73
12	97.2%	86.73	6.70	15.7%	2.36
20	97.2%	84.93	7.30	11.4%	2.18

Finding 4: Deadline pressure accelerates concession. Under the 6-round horizon, 82.4% of all deals close in the last two rounds, compared to 15.7% under 12 rounds and 11.4% under 20 rounds. The tight-deadline condition also produces the fastest concession rate (2.73 \$/round vs. 2.18 \$/round under 20 rounds), a 25% increase. H-2a is supported. The result is directionally consistent with deadline effects documented in human negotiation [13].

Finding 5: Prices are stable across horizon conditions. Mean deal prices differ by less than \$2 across conditions (\$84.93 to \$86.73), indicating that deadline proximity accelerates settlement timing without materially changing the final price. Agents concede faster but do not concede more: the surplus split remains approximately 83–88% buyer-favourable across all three conditions. This is consistent with the asymmetric concession pattern observed in Experiment 1: buyers absorb most of the concession pressure, but even under tight deadlines the seller price remains close to cost. The price stability across deadline conditions constitutes a theoretically interesting null result. In human deadline studies [13], deadline pressure typically affects both the timing of concessions and the magnitude of final concessions, with tighter deadlines producing lower-quality outcomes for one or both parties. The LLM result dissociates these two effects: timing is sensitive to the horizon (reflected in the last-two-round share shifting from 11.4% to 82.4%), but price is not. A plausible interpretation is that the LLM agent treats deadline salience as a “time to close” signal—an instruction to accelerate agreement—rather than as a “concede more” signal that would alter its positional target. This dissociation is a specific, testable behavioural difference between LLM and human negotiators that could be probed further through targeted deadline studies with human baselines.

6.3 Macro-Level Results: Market Outcomes and Mechanism Effects

Experiments 3, 4, 6, the Experiment 4 extension, and 5 shift the unit of analysis from individual sessions to multi-tick markets. All use `llm.free.language` agents with the gate disabled and 3 seeds.

6.3.1 Experiment 3: Market Dynamics

Experiment 3 observes a single-condition 10-tick market (10 pairs per tick, 3 seeds = 300 sessions total) to examine whether bilateral LLM negotiation produces stable aggregate price levels and whether within-tick price dispersion persists. The overall deal rate is 82.3% (247/300); the timeout rate is 9.7%.

Table 4: Market dynamics tick-level statistics averaged across seeds (300 sessions, 3 seeds). CV is coefficient of variation of mean price across ticks.

Tick	Mean price (\$)	Price std (\$)	Liquidity	Notes
0	83.68	14.54	—	Early
1	97.29	17.53	—	
5	92.16	5.51	—	Mid
9	84.50	16.51	—	Late
<i>Aggregate (all 10 ticks)</i>				
Mean price (all deals)	~\$89		—	—
Mean liquidity	—	—	0.81	High
Price trend slope	−0.60 per tick		—	Mild downward drift

Finding 6: Prices do not converge; dispersion persists throughout.

The coefficient of variation of tick-level mean prices fluctuates between 0.06 and 0.20 across the 10-tick window without a consistent downward trend. The market does not settle on a common price level, consistent with the structural property of this market design: each pair negotiates independently under private constraints, and no mechanism forces prices to converge across pairs. Within-tick price standard deviation ranges from \$5.51 to \$17.53, confirming that the Law of One Price fails in this market structure. H-3a is not supported: mean price does not stabilise within the observed horizon. H-3b is supported: within-tick dispersion remains substantial throughout. Importantly, the absence of price convergence is not a failure of the market or the agents—it is the theoretically predicted outcome for a system of independent bilateral negotiations. In such a structure, no price-discovery mechanism connects the outcomes of simultaneous pairs; each negotiation is resolved privately, producing a distribution of outcomes rather than a single equilibrium price. The observed persistent dispersion should therefore be understood as positive structural validation: the simulator faithfully instantiates the expected property of decentralised bilateral bargaining, rather than inadvertently collapsing to a degenerate uniform outcome.

Finding 7: Liquidity is high. Mean liquidity of 0.81 (81%) indicates that the large majority of matched pairs reach agreement, suggesting that the Qwen2.5-14B model is effective at extracting deals within 10 rounds under the given parameter distributions without gate assistance.

Finding 8: Surplus is approximately symmetric. Mean buyer surplus and seller surplus are approximately balanced (\$31.48 and \$31.68 respectively), consistent with the scenario parameters (buyer value from [\$80, \$150], seller cost from [\$50, \$120], creating varied ZOPA distributions). H-3c is not supported: the first-mover advantage observed in fixed-scenario sessions (Experiment 1) does not translate into asymmetric aggregate surplus in the market setting with heterogeneous parameters. This finding is in direct contrast to the 5.65:1 buyer-to-seller concession ratio observed in Experiment 1, and the contrast is itself informative. A plausible explanation is that under heterogeneous parameter draws, the randomness of the value and cost distributions dominates any systematic role-based asymmetry: in some pairings the buyer holds most of the bargaining leverage, in others the seller does, and these effects wash out in the aggregate. The buyer-first concession advantage observed under fixed-scenario conditions is therefore a property of that specific parameterisation, not a robust feature of the protocol when agents negotiate under varied private constraints.

6.3.2 Experiment 4: Stochastic Shock Response

Experiment 4 applies stochastic multiplicative shocks (30% per-tick probability; demand and supply multipliers drawn uniformly from [0.7, 1.3]) to a 10-tick market and compares outcomes against an unshocked baseline. 600 sessions were run (300 per condition, 3 seeds). Deal rate is 70.5% (423/600).

Table 5: Shock response: difference-in-differences (DiD) analysis (300 sessions per condition, 3 seeds). Pre-shock period: ticks 0–4; post-shock period: ticks 5–9. DiD is a structured descriptive comparison, not a causal estimate.

Condition	Pre-period mean price (\$)	Post-period mean price (\$)	Within-condition diff (\$)	DiD (\$)
no_shock	107.88	110.62	+2.75	
with_shock	111.44	107.66	−3.78	−6.53

Liquidity DiD: −0.060 (marginal suppression of deal rates under shocks).

Finding 9: Shocks produce a price-depressing effect of approximately \$6.53. The structured descriptive DiD comparison yields −\$6.53: the shock condition declines by \$3.78 across the two halves while the no-shock condition rises by \$2.75. This is directionally consistent with the prediction that mixed-direction stochastic shocks depress average prices and reduce liquidity, since negative shocks (demand multiplier <1 or supply multiplier >1) compress the ZOPA and reduce the floor available for deal completion. H-4b is supported directionally.

Finding 10: No consistent price recovery within the 10-tick window. Seed-level trajectories show heterogeneous behaviour: seed 42 exhibits partial recovery after the shock period, seed 123 shows continued decline, and seed 456

shows mild depression without recovery. There is no consistent recovery pattern. This finding should be treated as suggestive rather than conclusive: with only three seeds and 10 ticks each, the seed-to-seed variance dominates the aggregate pattern.

Finding 11: Liquidity impact is marginal. The DiD for liquidity is -0.060 , a modest reduction compared to the price effect. Stochastic shocks of symmetric magnitude (multipliers in $[0.7, 1.3]$ with mean 1.0) produce only small net liquidity effects on average, since positive and negative shocks partially offset. Tail events (large negative shocks) will produce locally severe liquidity drops within individual ticks, but these are diluted in the aggregate. H-4a is supported directionally: the shock condition shows greater tick-to-tick price variance than baseline. H-4c (asymmetric tail risk) cannot be evaluated with only three seeds, as within-seed variance dominates any tail analysis.

Interpretive caveat. The DiD comparison is a structured descriptive analysis. It is not a causal estimate: the pre- and post-periods coincide with a time trend in the no-shock condition ($+\$2.75$), and the shock condition’s evolution is influenced by the specific shock realisations drawn under each seed. The $\$6.53$ estimate should be understood as a directional signal under these particular parameter settings, not as a precise measure of shock impact. A post-hoc pooled comparison (Section 6.3.3) reports a different quantity—the mean price difference between shock and no-shock sessions pooled across all ticks—and is not contradictory to this DiD estimate.

6.3.3 Experiment 4 Extension: Anchor Environment

The Experiment 4 anchor-environment extension tests whether the item reference price in the agent prompt functions as an environmental price anchor that shapes both price levels and shock responsiveness. Three anchor modes—`fixed_anchor` (constant reference price, the default from Experiment 4), `updated_anchor` (dynamically updated to the prior tick’s mean deal price), and `no_anchor` (reference price omitted)—are each crossed with the shock/no-shock condition from Experiment 4. The `fixed_anchor` condition reuses the Experiment 4 dataset; the other two modes add 1,200 new sessions.

Finding 12: The reference price signal is a primary determinant of price levels. Across the three anchor modes, baseline mean prices span from approximately $\$79$ to $\$109$ —a range of roughly $\$30$ —despite identical buyer value and seller cost distributions in all conditions. H-4ext-a is strongly supported: the environmental reference signal shapes price levels substantially, exceeding the $\$4.63$ price response to a $\$20$ demand shock reported in Experiment 6 (Section 6.3.4). This finding demonstrates that LLM agents are highly sensitive to the informational environment in which they negotiate, not only to their private economic constraints. The $\$30$ spread is a strong result: it means that under these experimental conditions, the informational environment explains more

Table 6: Experiment 4 extension: informational anchor environment results. Mean price and liquidity are reported for the no-shock condition; shock price effect is the difference in mean deal price between the with-shock and no-shock conditions within each anchor mode. Fixed-anchor values are derived from the Experiment 4 dataset.

Anchor mode	Mean price, no-shock (\$)	Price std (\$)	Liquidity	Shock price effect (\$)
<code>fixed_anchor</code>	≈ 109	≈ 15	0.69	-6.53
<code>updated_anchor</code>	78.84	14.16	0.713	+4.15
<code>no_anchor</code>	99.21	7.73	0.867	-0.20

price variance than the underlying economic fundamentals do. This has a direct methodological implication: any LLM-based market simulation that embeds a reference price in the agent prompt is, intentionally or not, pre-determining a substantial portion of the price level. The reference signal is not a neutral contextual detail—it is a first-order design variable. This finding also helps explain the price stickiness observed in Experiment 6. If agents anchor to a prompt-embedded reference price rather than independently reasoning from their private constraints, then a \$20 shift in those constraints will produce only a muted price response: the agent continues to negotiate toward the anchored reference level rather than fully internalising the changed fundamental. The low elasticities in Experiment 6 (0.275 and 0.035) are thus plausibly a downstream consequence of the reference-price anchoring mechanism identified here.

Finding 13: Updated anchors create self-reinforcing price drift. Under `updated_anchor`, baseline mean prices settle at \$78.84, well below both the `fixed_anchor` ($\sim \$109$) and `no_anchor` (\$99.21) conditions. The mechanism is a self-reinforcing feedback loop: below-equilibrium deal prices in early ticks feed a lower reference price into subsequent prompts, which in turn produces lower opening offers and sustains further below-mean settlements. The shock price effect under this mode is *positive* (+\$4.15), reversing the sign observed in the fixed-anchor condition (-\$6.53). A plausible behavioural interpretation is that stochastic shocks partially disrupt the downward feedback mechanism, producing temporarily higher reference prices that lift the overall mean. The sign reversal cautions against treating the direction of the shock effect as a robust property of LLM negotiation behaviour independent of the informational anchor environment.

Finding 14: Removing the anchor stabilises prices but eliminates shock sensitivity. The `no_anchor` condition produces the most stable price series (std = \$7.73 vs. \$14.16 under updated-anchor) and the highest liquidity (0.867 vs. 0.69–0.71). However, the shock price effect is effectively zero (-\$0.20), indicating that without a reference signal, agents default to an internal prior and do not update it in response to fundamental changes. H-4ext-b is supported:

removing the anchor eliminates shock sensitivity.

Supplementary statistical check: pooled anchor-mode comparison.

The three anchor-mode conditions were compared using a pooled post-hoc analysis of 1,388 completed deals drawn from Experiment 4 (fixed-anchor arm, both shock conditions) and the Experiment 4 extension (updated-anchor and no-anchor arms, both shock conditions) across three seeds. Fixed-anchor data originate from Experiment 4; updated-anchor and no-anchor data from the Experiment 4 extension. The analysis uses Welch’s t -tests with Bonferroni correction for three pairwise comparisons (corrected $\alpha = 0.0167$). This is a post-hoc descriptive analysis, not a pre-registered confirmatory test. An important methodological caveat applies: sessions within a seed share market context, violating the independence assumption underlying the standard error estimates; consequently, reported standard errors may be underestimated and p -values should be interpreted with commensurate caution.

Under the tested model-prompt-protocol configuration, all three pairwise price-level comparisons are statistically significant after Bonferroni correction (Table 7). Effect sizes range from medium to large by conventional benchmarks ($d = 0.48$ to $d = 1.32$), consistent with the qualitative price-level ordering observed in Table 6.

Table 7: Post-hoc pairwise comparison of mean transaction prices across anchor modes (1,388 deals pooled over shock conditions, Welch’s t -test with Bonferroni-corrected $\alpha = 0.0167$). Fixed-anchor data from Experiment 4; updated-anchor and no-anchor data from the Experiment 4 extension.

Comparison	Δ mean (\$)	95% CI (\$)	p	Cohen’s d
fixed vs. updated	+26.79	[+24.05, +29.53]	1.5×10^{-68}	+1.32
fixed vs. no_anchor	+10.01	[+7.35, +12.67]	3.6×10^{-13}	+0.48
updated vs. no_anchor	−16.78	[−19.33, −14.23]	3.8×10^{-35}	−0.83

Within-anchor-mode shock sensitivity was assessed by comparing pooled mean transaction prices between the with-shock and no-shock sessions for each mode. The observed differences are: fixed-anchor +\$0.60 ($p = 0.77$, n.s.), updated-anchor +\$4.07 ($p = 0.032$, $d = 0.21$), and no-anchor +\$0.18 ($p = 0.92$, n.s.). Only the updated-anchor condition shows a statistically significant mean price difference between shock conditions, though the effect size is small ($d = 0.21$) and should be treated with the same non-independence caveat noted above.

A reconciliation with Finding 9 (Section 6.3.2) is warranted. The pooled fixed-anchor shock effect of +\$0.60 (n.s.) does not contradict the DiD estimate of −\$6.53 reported there. The DiD measures temporal price *change* across a pre/post split (ticks 0–4 vs. 5–9), capturing within-condition dynamics over time. The pooled shock comparison reported here measures the overall mean price *level* difference between shock and no-shock sessions across all ticks, col-

lapsing the temporal dimension. Both statistics are valid descriptions of different aspects of the same data. Together, they suggest that the fixed-anchor shock effect manifests as a directional temporal shift within each run rather than as a persistent mean price elevation detectable in the session-level pooled comparison.

Behavioural interpretation. The \$30 price-level spread across anchor modes, combined with the sign reversal in the shock effect, indicates that LLM price-setting behaviour is strongly conditioned on the reference information presented in the prompt. Agents do not independently discover prices from private constraints alone; they use the environmental reference signal as an anchor around which to negotiate. This behaviour is consistent with the anchoring literature [16, 2], though the mechanism differs: in human studies, anchoring operates through an opponent’s first offer, whereas here it operates through a prompt-embedded environmental signal. For experimental design, this implies that anchor mode must be explicitly controlled and reported as a first-order design variable.

6.3.4 Experiment 6: Structural Supply-Demand Shifts

Experiment 6 applies deterministic, directional \$20 shifts to buyer values (demand shock) or seller costs (supply shock) and compares outcomes against a baseline. 576 sessions were run (192 per condition, 3 seeds). Deal rate is 70.5% (406/576).

Welfare accounting identity caveat. Before reporting welfare figures, a structural caution is necessary. Total welfare in this simulator is defined as the sum of buyer surplus and seller surplus: $(v_b - p) + (p - c_s) = v_b - c_s$, where v_b is buyer value, c_s is seller cost, and p is the deal price. The price p cancels. Consequently, aggregate welfare is mechanically determined by the distribution of (v_b, c_s) pairs that reach deals, not by the price at which those deals close. Welfare comparisons across supply-demand conditions reflect changes in the composition of trading pairs, not agent-level bargaining behaviour. This identity must be borne in mind when evaluating the findings below. Price movements, by contrast, require genuine LLM behavioural adaptation and are therefore the more informative signal of agent price discovery.

Table 8: Supply-demand experiment outcomes by condition (192 sessions per condition, 3 seeds). Buyer value and seller cost columns show mean values across the distribution used in each condition.

Condition	Mean price (\$)	Liquidity	Total welfare (\$)	Buyer surplus (\$)	Seller surplus (\$)	Buy
baseline	109.33	0.724	65.60	21.40	44.19	
demand_shock	113.96	0.724	87.24	39.07	48.18	
supply_shock	110.50	0.667	49.84	22.16	27.67	

Table 9: Price elasticity of supply and demand shocks. Elasticity is defined as proportional price change divided by proportional fundamental change (a $\pm\$20$ shift on a mean base).

Shock	Input shift (\$)	Price change (\$)	Price change (%)	Elasticity
Demand shock	+20 (buyer value)	+4.63	+4.2%	0.275
Supply shock	+20 (seller cost)	+1.17	+1.1%	0.035

Finding 15: Prices exhibit strong stickiness under both shocks. Despite \$20 shifts in underlying fundamentals, transaction prices move by only \$4.63 under the demand shock (+4.2%) and \$1.17 under the supply shock (+1.1%). The implied price elasticities are 0.275 and 0.035 respectively—far below the full pass-through (elasticity of 1.0) that classical price theory would predict. Agents appear to anchor to a stable reference price range and adjust it only weakly in response to changing fundamentals, even when those fundamentals are encoded in their private constraints. This price stickiness is the most substantively important finding in Experiment 6, because it requires the LLM to update prices in response to changed fundamentals rather than following mechanically from the accounting identity.

Finding 16: Demand shock raises prices; supply shock barely affects them. The demand shock (+\$20 buyer value) produces a \$4.63 price increase, four times larger than the supply shock’s \$1.17 despite identical input magnitudes. H-6a is supported directionally: the demand shift raises mean price, though the magnitude is far below full pass-through. H-6b is partially supported: the supply shift raises prices but by a negligible amount (\$1.17), and liquidity drops from 0.724 to 0.667, consistent with ZOPA compression. This asymmetry is directionally inconsistent with classical competitive market predictions, in which a cost-floor increase (supply shock) should produce stronger price pass-through than a value increase (demand shock), because sellers can enforce a hard minimum floor at their cost. One possible explanation is that buyers drive pricing more aggressively than sellers in this bargaining protocol—consistent with the buyer’s first-mover advantage—so that demand-side changes propagate into prices more readily. An alternative explanation is model-specific anchoring to cost-adjacent prices. Neither explanation is resolvable without targeted ablation.

Finding 17: Welfare changes track the accounting identity. Total welfare rises by \$21.65 under the demand shock (\$65.60 to \$87.24) and falls by \$15.76 under the supply shock (\$65.60 to \$49.84). As noted in the caveat above, these changes are mechanically determined by the shift in $(v_b - c_s)$ distributions and are not evidence of agent-level welfare optimisation. Liquidity is unchanged under the demand shock (0.724 in both conditions), indicating that the wider value distribution does not translate into more deals—a further sign of price

stickiness, because buyers with higher values could afford to accept the seller’s price but do not consistently close faster.

Finding 18: Supply shock redistributes surplus toward buyers, relative to total. Under the supply shock, buyer surplus rises marginally (+\$0.76, from \$21.40 to \$22.16) while seller surplus falls sharply (−\$16.52, from \$44.19 to \$27.67). The buyer share of total welfare increases from 33% to 44%. This distributional shift is consistent with cost-floor pressure: higher seller costs reduce the available seller margin at the prices at which deals are closing, even though the closing price itself barely changes. Sellers absorb the cost shock with little ability to pass it forward through higher prices. H-6c is directionally supported: the liquidity drop under the supply shock (0.724 to 0.667) exceeds what ZOPA compression alone would predict, suggesting a behavioural component in which agents fail to reach agreement even when a narrow positive ZOPA exists.

6.3.5 Experiment 5: Matching Mechanism Comparison

Experiment 5 compares random matching (baseline) against surplus-maximising (oracle) matching in an 8-tick market. 335 sessions were run (random: 192; surplus-max: 143; the lower surplus-max count arises because the matcher excludes negative-ZOPA pairs). Overall deal rate is 83.9%.

Table 10: Mechanism comparison outcomes (3 seeds, 8 ticks per seed). Total surplus is the sum of buyer and seller surplus across all deals.

Metric	random	surplus_max	Difference
Total sessions	192	143	−49
Total surplus (\$)	49.48	53.05	+3.57 (+7.2%)
Price std (\$)	17.92	16.14	−1.78
Liquidity	0.85	0.82	−0.03
Mean buyer surplus (\$)	25.52	26.61	+1.09
Mean seller surplus (\$)	23.96	26.43	+2.47

Finding 19: Surplus-maximising matching improves total welfare by 7.2%. Total surplus rises from \$49.48 under random matching to \$53.05 under surplus-max matching, a 7.2% improvement. The surplus-max matcher achieves this by directing trades exclusively toward pairs with positive ZOPA overlap, increasing the probability that any given session produces a deal. H-5b is supported.

Finding 20: Mechanism design benefits both parties. Both buyer surplus (+\$1.09) and seller surplus (+\$2.47) increase under the improved mechanism. Seller surplus gains are proportionally larger. This pattern is consistent

with the scenario producing a balanced surplus environment, where sellers can negotiate outcomes closer to their preferred range.

Finding 21: Liquidity is marginally lower under surplus-max, yet welfare is higher. Liquidity falls slightly from 0.85 to 0.82 despite the mechanism filtering out low-ZOPA pairs. This counter-intuitive result may arise because surplus-max occasionally creates highly asymmetric pairings in which one side holds all the bargaining leverage, making agreement harder to reach even when ZOPA overlap exists. The net effect is positive for welfare because the deals that do close are of higher quality. H-5a (higher deal rate under surplus-max) is not supported; the mechanism improves welfare quality rather than quantity.

6.4 Cross-Experiment Synthesis

This section synthesises patterns across experiments and organises findings by evidence strength.

Theme 1: Asymmetric concession burden as a protocol-level pattern.

Across Experiments 1, 2, and 6, a consistent pattern emerges: buyers concede more than sellers. In Experiment 1, the buyer-to-seller concession ratio is 5.65:1, and all deals cluster in the bottom 28% of the ZOPA. In Experiment 2, buyer surplus exceeds 83% of the ZOPA across all deadline conditions, with the seller barely moving from its initial position. In Experiment 6, buyer surplus is lower than seller surplus at baseline (33% vs. 67% of total welfare), and the supply shock redistributes only modestly toward buyers. This asymmetry is consistent with the buyer-first turn order in the alternating-offer protocol and with the observed seller anchoring behaviour, but it is also sensitive to the specific parameter settings used. Whether this asymmetry is an intrinsic property of LLM bargaining or an artefact of the scenario configuration cannot be determined without ablation of the role-assignment and parameter settings.

Theme 2: Welfare as accounting identity versus price as behavioural signal.

A recurring interpretive challenge across Experiments 6 and 5 is that welfare measures partly reflect the accounting identity ($v_b - c_s$) rather than agent decision quality. Welfare increases when the distribution of (v_b, c_s) pairs shifts favourably, and when trades are concentrated in high-surplus pairs (as in surplus-max matching), regardless of how agents negotiate. Price movements, by contrast, require the LLM agent to adjust its offer in response to changed fundamentals. The consistently low price elasticities in Experiment 6 (0.275 for demand shocks, 0.035 for supply shocks) and the persistent within-tick price dispersion in Experiment 3 indicate that price discovery is weak: agents do not efficiently update prices in response to shifting values and costs. This is the more diagnostically useful finding about LLM bargaining capability. This distinction—between welfare metrics that track an accounting identity and price metrics that require genuine agent adaptation—should inform how future LLM

negotiation studies evaluate market outcomes: reporting welfare without separately reporting price elasticity risks attributing to agent capability what is in fact a structural property of the parameter design.

Table 11: Cross-experiment summary of findings, organised by evidence strength. “Strong” indicates a finding consistent across all three seeds with directional clarity; “Moderate” indicates a finding with directional support but high cross-seed variance or limited replication; “Weak” indicates a finding from a pattern that is unclear or not replicated.

Pattern	Experiments	Strength	Primary driver
Concave (Boulware) concession curves	1	Strong	Agent
Asymmetric concession burden (5.65:1)	1, 2	Strong	Protocol/Agent
Deadline acceleration (82% last-2 at 6 rounds)	2	Strong	Agent
Reference-signal price sensitivity (~\$30 spread)	4ext	Strong	Prompt/Design
Surplus-max mechanism $\rightarrow$ +7.2% welfare	5	Moderate	Mechanism
Shock price depression (DiD -6.53)	4	Moderate	Protocol/Agent
Anchor-mode moderation of shock sensitivity	4, 4ext	Moderate	Prompt/Design
Price stickiness (elasticity 0.03–0.28)	6	Moderate	Agent
Welfare tracks fundamentals (accounting identity)	6	(Mechanical)	Design
Price convergence in market	3	Weak	None observed
Consistent shock recovery	4	Weak	Not observed

Theme 3: Deal success rates vary systematically across experimental conditions. Deal success rates range from 97.2% in fixed-scenario experiments (Experiment 1) to 70.5% in distribution-mode market experiments (Experiments 4 and 6). Table 12 summarises deal rates across all experiments.

The primary determinant of deal success is the experimental setting: fixed-scenario experiments with wide ZOPAs (A, C) achieve near-complete agreement, while distribution-mode experiments produce lower rates because some randomly matched pairs have narrow or absent ZOPAs. Among market experiments, the `no_anchor` condition in the Experiment 4 extension achieves the highest deal rate (86.7%), suggesting that removing the reference price signal may reduce anchoring-induced rigidity that prevents agreement. The supply shock in Experiment 6 produces the lowest deal rate (66.7%), consistent with ZOPA compression from higher seller costs.

Theme 4: Comparison to human negotiation patterns. Several observed behavioural patterns are directionally consistent with patterns documented in the human negotiation literature, though all comparisons are indirect (no human participants were tested under the same protocol).

Deadline-driven concession acceleration. The Experiment 2 finding—82.4% of deals close in the final two rounds under a 6-round horizon, compared to 11.4% under 20 rounds—is directionally consistent with Roth et al.’s [13] documentation of deadline effects in human bargaining. However, LLM agents do

Table 12: Deal success rates across all experiments.

Exp	Name	Deal rate (%)	Sessions
1	Concession (LLM)	97.2	72
1	Concession (rule-based)	97.2	72
2	Deadline (6 rounds)	94.4	72
2	Deadline (12 rounds)	97.2	72
2	Deadline (20 rounds)	97.2	72
3	Market dynamics	82.3	300
4	Shock (no shock)	≈ 69	300
4	Shock (with shock)	≈ 72	300
6	Supply-demand (baseline)	72.4	192
6	Supply-demand (demand shock)	72.4	192
6	Supply-demand (supply shock)	66.7	192
4ext	Anchor (fixed, no shock)	≈ 69	300
4ext	Anchor (updated, no shock)	71.3	300
4ext	Anchor (no anchor, no shock)	86.7	300
5	Mechanism (random)	85.0	192
5	Mechanism (surplus-max)	82.0	143

not adjust final prices across deadline conditions (\$84.93 to \$86.73, a \$2 range), whereas human deadline studies typically document both timing and price effects. This dissociation—timing sensitivity without price sensitivity—is a specific, testable behavioural difference between LLM and human negotiators. It suggests that the LLM responds to deadline salience primarily as a coordination signal (“the window is closing; close the deal now”) rather than as a pressure signal that alters its assessment of an acceptable price. Whether this difference is model-specific, prompt-specific, or a general property of LLM bargaining is a question that could be probed directly through studies that vary both the deadline condition and the human baseline under the same protocol.

Boulware-style concession. The concave concession trajectory observed in Experiment 1 (large initial concession followed by rapid hardening) is consistent with the Boulware bargaining pattern documented in the negotiation literature [12]. The prompt instructs monotonic concession direction but does not dictate step size or timing; the frontloaded concession pattern is therefore partly attributable to LLM behaviour, though the boundary between prompt-induced and agent-driven dynamics cannot be determined from current experiments.

Anchor sensitivity. The Experiment 4 extension demonstrates that LLM agents are strongly conditioned on the environmental reference price, consistent with the anchoring literature [16, 2], though the mechanism differs: in human studies, anchoring operates through an opponent’s first offer, whereas here it operates through a prompt-embedded environmental signal.

Protocol-level considerations. Across all experiments, the bargaining protocol is held constant: agents operate under a finite-horizon counteroffer protocol with structured action extraction, monotonic concession instructions, and

pre-computed feasibility information. These protocol choices constrain concession direction and steer toward deal finalization when offers are feasible. The consistently low seller concession observed in Experiment 1 may partly reflect prompt asymmetry. Prompt-level ablation studies—testing deal rates and concession dynamics with and without the monotonic concession instruction, the accept-condition block, and the reference price signal—would further clarify the boundary between agent-driven and protocol-induced behaviour.

7 Discussion

This section interprets the experimental results, examines whether LLM agents produce behaviourally recognisable negotiation patterns, identifies the principal methodological tensions, situates the findings relative to prior work, and identifies the limitations of the current study.

7.1 Behavioural Findings: What the Experiments Reveal About LLM Negotiation

The experimental programme reveals a consistent set of behavioural patterns that emerge from LLM-driven negotiation without gate-driven enforcement (all acceptances across the programme are LLM-driven; prompt steering remains active as an uncontrolled variable).

Concession and bargaining dynamics. LLM agents produce concave (Boulware-style) concession trajectories (Finding 1): the buyer makes a large opening concession of \$13.31 between rounds 0 and 2, then hardens rapidly. The seller exhibits strong anchor rigidity, barely moving from its initial position (Finding 3). The resulting 5.65:1 buyer-to-seller concession ratio indicates an asymmetric bargaining dynamic in which the buyer bears the concession burden. Deadline effects are clearly visible: under the 6-round horizon, 82.4% of all deals close in the final two rounds, with a 25% higher concession rate compared to the 20-round horizon (Finding 4).

Deal success and market dynamics. Deal success rates vary systematically from 97.2% in fixed-scenario experiments to 66.7–72.4% in distribution-mode market experiments, primarily reflecting ZOPA availability. Market liquidity of 81% (Experiment 3) indicates that the Qwen2.5-14B model is effective at reaching deals within 10 rounds without gate assistance. Prices exhibit persistent within-tick dispersion without convergence (Experiment 3), and directionally correct but substantially attenuated responses to supply and demand shocks (Experiments 4 and 6).

Informational anchor sensitivity. the Experiment 4 extension demonstrates that the environmental reference price in the agent prompt is a primary determinant of price levels: mean transaction prices span approximately \$30 across

anchor modes under identical fundamentals. This sensitivity extends to shock responsiveness, with the shock effect sign reversing between fixed-anchor and updated-anchor modes. This finding establishes that LLM agents do not independently discover prices from private constraints alone; the informational environment materially shapes bargaining outcomes.

Micro–macro capability gap. A central insight running across the experimental programme is that LLM agents demonstrate separable capabilities in deal-closing and price-discovery. Deal-closing performance is high: market liquidity reaches 81% without gate assistance, and individual sessions settle reliably within the round limit. Price-discovery performance is low: price elasticities to fundamental shocks are 0.035–0.275, and prices track the prompt-level reference signal rather than the agents’ private constraints. This dissociation—strong convergence to agreement, weak convergence to informationally correct prices—is the primary structural finding of the project and organises the discussion in Sections 7.3 and 7.4.

Institutional mechanism effects. Surplus-maximising matching improves total welfare by 7.2% relative to random matching (Experiment 5), with gains shared across both buyers and sellers. This demonstrates that mechanism design can improve market outcomes independently of agent sophistication.

7.2 LLM Agents and Human Negotiation Behaviour

A central question motivating this project is whether LLM agents produce negotiation behaviour that is recognisably similar to patterns documented in human negotiation studies. The experimental programme enables directional comparison against three well-established human bargaining patterns, though all comparisons are indirect: no human participants were tested under the same protocol.

Where patterns align. Deadline-driven concession acceleration (Experiment 2) is directionally consistent with Roth et al.’s [13] documentation of deadline effects in human bargaining: agents make disproportionately large concessions near the deadline under time pressure. The concave concession trajectory (Experiment 1) is consistent with the Boulware bargaining pattern [12] documented in the negotiation literature, in which a negotiator makes early accommodations but becomes increasingly resistant. Reference-price sensitivity (the Experiment 4 extension) parallels the first-offer anchoring effect [2], though through a different mechanism: environmental prompt-level anchoring rather than opponent-generated first offers.

Where patterns diverge. Prices do not shift meaningfully with deadlines (Experiment 2, Finding 5): mean deal prices differ by less than \$2 across horizon conditions, whereas human deadline studies typically document both timing

and price effects. This dissociation between timing sensitivity and price sensitivity under deadline conditions is a specific, testable behavioural difference that distinguishes LLM negotiation from human negotiation. In human studies, deadline pressure typically induces larger concessions—affecting both when and at what price deals close; LLM agents show only the former effect, suggesting they respond to deadline salience as a procedural cue to close the deal rather than as strategic pressure to revise their price target. Under the finite-horizon counteroffer protocol, agents do not voluntarily walk away from profitable sessions; the round limit serves as the outside option rather than an active rejection decision. This protocol eliminates the threat-of-rejection dynamic that is central to strategic human negotiation behaviour. Price stickiness (Experiment 6) indicates that agents do not efficiently update prices in response to changing fundamentals, a form of departure from the adaptive price-discovery behaviour expected of experienced human negotiators.

Conservative interpretation. The observed alignments are consistent with LLM agents having absorbed negotiation-relevant behavioural patterns from training data, but this cannot be confirmed from the current experiments. The appropriate framing is that LLM agents exhibit qualitative behavioural patterns that are interpretable through the same theoretical lens as human negotiation behaviour, without claiming that the underlying cognitive or strategic mechanisms are similar. Direct human comparison under identical protocol would be required to substantiate stronger claims.

7.3 Price Stickiness and Weak Price Discovery

A substantively important finding is the systematic failure of LLM agents to pass through fundamental changes into prices. In Experiment 6, a \$20 shift in buyer value produces only a \$4.63 price increase (elasticity 0.275), and a \$20 shift in seller cost produces only a \$1.17 price increase (elasticity 0.035). Agents adjust prices only weakly in response to changing fundamentals, even when those fundamentals are encoded in their private constraints. the Experiment 4 extension identifies the prompt-level reference price as a substantial contributor, though the full mechanism is not isolated. This price stickiness is observed in sessions where all settlements are LLM-driven and the gate is absent. The parallel finding in Experiment 3—persistent within-tick price dispersion without convergence over 10 ticks—reinforces the same conclusion: under the tested model–prompt combination, agents function as effective deal-closing mechanisms but poor price-discovery mechanisms.

the Experiment 4 extension (Section 6.3.3) provides supporting evidence that the prompt-level reference price is one contributor to price stickiness. Across the full range of anchor modes, mean transaction prices vary by approximately \$30 (from \$79 under `updated_anchor` to \$109 under `fixed_anchor`), indicating that the environmental reference signal influences price levels substantially. A post-hoc pooled comparison across 1,388 deals (Section 6.3.3) confirms that

all three pairwise price differences are statistically distinguishable after Bonferroni correction, with effect sizes ranging from $d = 0.48$ (fixed vs. no-anchor) to $d = 1.32$ (fixed vs. updated), indicating that anchor-mode effects are practically as well as statistically meaningful under this protocol. The `no_anchor` condition also eliminates shock sensitivity almost entirely (shock price effect of $-\$0.20$), indicating that without an external reference signal, agents default to an internal prior and do not adjust prices in response to fundamental changes. The `updated_anchor` condition, which feeds prior deal prices back into the prompt, produces a self-reinforcing downward drift and reverses the sign of the shock effect. The pooled shock-sensitivity analysis additionally shows that the updated-anchor condition is the only one that produces a statistically significant mean price difference between shock and no-shock sessions ($p = 0.032$, $d = 0.21$); fixed-anchor and no-anchor conditions show no significant pooled shock effect ($p = 0.77$ and $p = 0.92$ respectively). This result is consistent with Findings 13 and 14 but should be interpreted cautiously: the updated-anchor shock effect is marginal ($d = 0.21$, a small effect by conventional benchmarks) and the analysis is post-hoc with sessions clustered within seeds. These findings suggest that price stickiness in this simulator is not solely a property of the LLM’s reasoning limitations but is partly a consequence of the environmental information structure: agents are sensitive to the reference the prompt provides. The full mechanism underlying price stickiness—whether it arises primarily from training-data priors, prompt-level anchoring, or the conservative opening strategy observed in Experiment 1—is not resolved by the current experiments. This limitation also has a practical dimension: deployed LLM negotiation systems using similar prompt architectures may exhibit systematic mis-pricing that does not self-correct as market conditions change, because agents anchor to the reference price embedded in the prompt rather than adapting to the informational content of offers received.

Price stickiness also has an important implication for interpreting welfare results. As noted in Section 6.3.4, total welfare in this simulator is defined as $(v_b - c_s)$, which is independent of the deal price p . Welfare comparisons across supply-demand conditions therefore reflect shifts in the composition of trading pairs, not agent-level price adjustment. Welfare results look strong because they are partly mechanical; price results look weak because they require genuine LLM behavioural adaptation. This distinction is the most important interpretive caution for Experiments 6 and 5.

Protocol design choices. The bargaining protocol implements a finite-horizon counteroffer format: agents counter-offer or accept, and episodes that reach the round limit end as no-deal outcomes. The prompt additionally instructs agents to concede monotonically in the correct direction (buyers upward, sellers downward). Both choices were introduced for practical reasons: without structured deal finalization, small language models frequently terminate profitable negotiations prematurely, producing near-zero deal rates; without directional concession guidance, agents sometimes concede in the wrong direction. The concession

magnitude and *timing* remain interpretable (the instruction constrains direction, not step size), but the monotonic concession pattern reported in Experiment 1 cannot be fully attributed to emergent LLM strategy. This illustrates a recurring tension between engineering reliability and experimental cleanliness.

Demand-side and supply-side asymmetry. Experiment 6 identifies an asymmetry in the direction opposite to what classical theory predicts: the demand shock raises prices by \$4.63 (+4.2%) while the supply shock raises them by only \$1.17 (+1.1%), despite identical \$20 input magnitudes. Standard supply-demand intuition suggests that a hard seller cost floor should produce stronger pass-through than a buyer value increase, but the observed result reverses this expectation. A plausible explanation is that buyers, as first movers, anchor prices more strongly than sellers: LLM buyers open higher under a demand shock because their value ceiling rises, but LLM sellers do not commensurately raise their counter-offers under a supply shock. Alternative explanations include the conservative buyer opening strategy documented in Experiment 1 (where free-language buyers open at \$71.88 against a \$120 value) and model-specific anchoring to a fixed reference range. None of these explanations can be separated without targeted ablation.

The supply shock also reduces market liquidity from 0.724 to 0.667, consistent with a tighter ZOPA degrading LLM convergence even when agreement is structurally possible.

Prompt architecture as a methodological consideration. In human negotiation research, the cognitive architecture of the bargainer is relatively fixed; experimental variation comes from the task, the opponent, and the information structure. In LLM negotiation, the prompt shapes the agent’s decision process, and changes to prompt design—adding a reasoning scaffold, including or excluding surplus information, steering toward or away from rejection—can alter outcomes. LLM negotiation results should therefore be understood as characterising a specific model–prompt–protocol combination rather than LLM capability in general. The Experiment 1 comparison between rule-based and free-language agents shows that the decision architecture shapes offer behaviour: the free-language agent produces concave concession trajectories while the rule-based agent follows a fixed linear schedule. Whether this reflects emergent strategic reasoning or prompt-induced output distribution cannot be determined from the current experiments.

7.4 Micro–Macro Tension

A structural tension runs through the experimental programme: designs that preserve raw agent behaviour may weaken aggregate market regularity, while stronger enforcement can stabilise outcomes but partially replaces agent decision-making.

As described in Section 4.3.2, the simulator includes a feasibility gate that, when enabled, overrides agent decisions by automatically accepting feasible offers. Preliminary runs with the gate enabled showed substantial gate involvement, producing near-100% deal rates in wide-ZOPA conditions but simultaneously suppressing the very dynamics the experiments were designed to study. With the gate disabled, two mediation layers remain active between the model’s generation and the enacted action: the rethink loop corrects format and self-constraint violations inside the agent, and the ActionJudge enforces protocol and global-bound constraints at the session level.

With the gate off, agents produce concave concession trajectories (Finding 1) and deadline-sensitive timing (Finding 4)—patterns that preliminary gate-on runs showed were suppressed by enforcement. Market liquidity is 81% with zero gate involvement (Finding 7), indicating that the Qwen2.5-14B model can sustain a functioning market without enforcement assistance. The micro–macro tension therefore manifests differently: enforcement is not needed to make the market function, but the absence of enforcement exposes a different limitation—price stickiness.

The price stickiness finding (elasticities 0.03–0.28, Finding 12) is the clearest example of this tension. Individual agents close deals effectively (micro-level function), but they do not update their price targets efficiently in response to changing fundamentals. The result at the market level is persistent price inertia: the market clears at prices that reflect the agents’ anchored reference range rather than the shifted equilibrium implied by the new parameter distributions. Effective deal closure (micro) and effective price discovery (macro) appear to be separable capabilities, with LLM agents demonstrating the former more reliably than the latter.

7.5 Comparison to Prior Work

We compare our findings to the prior work reviewed in Section 2.

Generative agent simulations. Park et al. [10] demonstrated that LLM agents exhibit emergent social behaviours in open-ended environments. Our market experiments (3 and 4) apply a related approach to an economic setting with hard structural constraints: LLM agents produce aggregate patterns—persistent price dispersion, varying liquidity, and surplus asymmetry—without being explicitly programmed to produce these outcomes. The key difference is that our environment is constrained (hard budget and cost limits, structured protocol) rather than open-ended, enabling direct comparison against theoretical predictions.

LLM negotiation. Abdelnabi et al. [1] find that deliberation prompting improves agreement quality in multi-party, multi-issue negotiation games. Our Experiment 1 provides a complementary finding in a simpler setting: the free-language agent produces concave concession trajectories with a 5.65:1 buyer-to-seller concession ratio, illustrating how decision architecture shapes bargaining

dynamics even in a single-issue bilateral protocol. Our setting is single-issue bilateral bargaining, which isolates price dynamics from the issue-trading effects present in their multi-issue design.

Mao et al. [9] show that personality-assigned prompts shift negotiation outcomes. Our prompt architecture (acceptance hints, concession-direction instructions) functions similarly as a prompt-level modifier; the methodological difference is that we hold prompt architecture constant as a fixed protocol parameter rather than studying it as an independent variable. All reported experiments use the default neutral communication strategy.

Lewis et al. [7] found that end-to-end trained negotiation agents develop non-human-like strategies, including degenerate repetitive behaviour. Our LLM agents show a different form of departure from human norms: rather than degenerating, they fail to efficiently update prices in response to changing fundamentals while maintaining otherwise reasonable bargaining behaviour. Both findings suggest that artificial agents do not spontaneously replicate the full spectrum of human negotiation behaviour, though the underlying mechanisms differ—reinforcement learning via self-play versus prompt-conditioned inference.

Classical theory. Several findings are directionally consistent with qualitative predictions from bilateral bargaining theory. Market prices exhibit persistent within-tick dispersion consistent with independently negotiated bilateral pairs [15], and the buyer-first protocol produces buyer-favourable surplus splits consistent with first-mover advantage [14]. Experiment 6 shows that a +\$20 demand shock raises prices by \$4.63 (+4.2%) and a +\$20 supply shock raises them by \$1.17 (+1.1%), both directionally consistent with competitive market predictions but substantially attenuated.

The deadline result is directionally consistent with Roth et al. [13]: concession rates are 25% higher under the 6-round horizon and 82.4% of deals close in the final two rounds. However, one finding runs counter to classical supply-demand theory: the demand shock produces a \$4.63 price increase (elasticity 0.275) while the symmetric supply shock produces only a \$1.17 increase (elasticity 0.035). Standard theory predicts that a binding seller cost floor should produce at least as strong a price pass-through as a buyer value increase of the same magnitude; the observed result is the reverse. This demand-supply price asymmetry is therefore a substantive negative finding, not merely an attenuated version of the classical prediction.

AgenticPay and open-weight model limitations. Liu et al. [8] introduce AgenticPay, a benchmark for multi-agent buyer–seller negotiation driven by natural language, spanning bilateral to many-to-many settings. Their evaluation of open-weight models on 111 negotiation tasks provides an informative external reference for interpreting our results. On the AgenticPay benchmark, Qwen3-14B—a model of the same nominal parameter scale (14B) as our Qwen/Qwen2.5-14B-Instruct, though from a different model generation—achieves a GlobalScore of 63.9 with a 20.7% timeout rate, compared to 100%

deal rates and GlobalScores above 81 for frontier proprietary models. Our own results show a 9.7% timeout rate in the market dynamics experiment and price elasticities of 0.03–0.28, consistent with the AgenticPay finding that open-weight models engage in price discovery but struggle with long-horizon strategic convergence. The AgenticPay benchmark provides an illustrative external reference, though the experimental tasks, protocols, and model generations differ substantially; no direct performance comparison between Qwen3-14B and Qwen2.5-14B-Instruct is intended. Both studies observe that models of comparable but not identical specification handle the mechanics of bilateral bargaining while showing systematic limitations in price convergence, but the underlying mechanisms are not established by either study.

7.6 Limitations

We identify nine limitations of the current study, ordered by their impact on the interpretation of results.

1. **Single model family; no cross-model comparison.** All experiments use `Qwen/Qwen2.5-14B-Instruct` at temperature 0.2. Results may not generalise to other model families, scales, or temperatures. In particular, the emergence of concave concession curves and deadline effects may reflect model-specific properties rather than general properties of LLM bargaining. No cross-model comparison is provided.
2. **Price stickiness: finding and limitation simultaneously.** The low price elasticities in Experiment 6 (0.035–0.275) are a substantively important finding, but they are also a limitation: agents do not discover prices efficiently, meaning the simulator cannot be used to test price discovery mechanisms that require agents to update prices in response to changing fundamentals. The mechanism underlying stickiness—whether it arises from training-data anchoring, prompt-level reference framing, or the conservative opening strategy observed in Experiment 1—is not resolved by the current experiments. At the same time, the finding itself is informative: it reveals that LLM agents are more responsive to environmental reference information embedded in the prompt than to private constraint information encoded in their own parameters, a distinction with implications for how such agents process strategic information.
3. **No prompt ablation.** The accept-condition block, monotonic concession instructions, the counteroffer-only action space, and deadline-salience language are all active across experiments but none are individually ablated. Their separate effects on deal rates, concession shape, price stickiness, and deadline sensitivity are unknown. The monotonic concession pattern (Finding 1) is partly attributable to the directional concession instruction; the seller’s observed resistance to concession (Finding 3) may partly reflect prompt asymmetry rather than intrinsic role-based behaviour.

4. **Three seeds is limited replication.** All experiments use three seeds (42, 123, 456). This provides basic cross-seed consistency checks and directional replication, but it is insufficient for hypothesis testing or effect-size estimation. Seed-level variance is visible in the shock recovery experiment (Experiment 4), where individual seeds show divergent trajectories. Most analyses are descriptive (means, standard deviations). The post-hoc pooled anchor-mode comparison reported in Section 6.3.3 constitutes a partial exception: it applies Welch’s t -tests with Bonferroni correction to 1,388 pooled deal observations. However, this analysis increases the number of observations rather than the number of independent seeds (still 3), and sessions within a seed share market context, meaning standard errors from session-level t -tests may be underestimated; the analysis should be treated as indicative rather than as a formally powered test.
5. **No human baseline.** All experiments compare LLM agents against rule-based or other LLM agents. No human negotiation data is collected under the same protocol. Claims about departure from or alignment with human negotiation behaviour therefore rely on the external literature rather than direct comparison.
6. **Single-price negotiation.** All sessions negotiate a single price. Real markets involve quality, delivery terms, and bundling. Multi-issue negotiation may produce qualitatively different dynamics, including the logrolling and issue-trading effects studied by Lewis et al. [7].
7. **No cross-run learning.** Agents do not accumulate experience across sessions within the completed experiments. The memory and reputation agents are implemented but untested. Whether LLM agents adapt strategies over repeated interactions remains an open question.
8. **LLM non-determinism.** Even at temperature 0.2, model outputs are not fully deterministic. The seeded RNG controls market-level randomness (population generation, matching, shocks) but not LLM inference randomness. This introduces uncontrolled variance into all LLM-agent conditions.
9. **Demand-supply price asymmetry mechanism unresolved.** The demand shock produces a \$4.63 price increase (elasticity 0.275) while the supply shock produces only \$1.17 (elasticity 0.035) despite identical \$20 input magnitudes. Three candidate explanations—buyer first-mover dominance, cost-floor anchoring by sellers, and nonlinear ZOPA-width sensitivity—are consistent with the data but have not been isolated through targeted ablation.

8 Conclusion and Future Work

8.1 Summary

This thesis set out to investigate whether LLM agents produce economically interpretable negotiation behaviour in a structured bilateral market, and to examine how simulator design choices—feasibility enforcement, prompt architecture, information structure—shape the outcomes that experiments can observe and attribute to agents.

The central empirical finding is a micro–macro capability gap: LLM agents are effective deal-closers but poor price-discovery mechanisms. Market liquidity reaches 81% without gate assistance, individual sessions settle reliably within the round limit, and deal success rates range from 97.2% in fixed-scenario experiments to 66.7% under adverse market conditions. Yet price elasticities to fundamental shocks are only 0.035–0.275, and the Experiment 4 extension shows that mean transaction prices span approximately \$30 across anchor modes under identical fundamentals. Agents anchor to the reference price signal in the prompt rather than to their private constraints, producing strong deal convergence at informationally inaccurate prices.

Supporting this finding, Experiments 1 and 2 document concave (Boulware-style) concession trajectories with a 5.65:1 buyer-to-seller concession ratio and deadline-driven settlement acceleration (82.4% of deals in the final two rounds; 25% higher concession rate under a six-round horizon). These micro-level patterns are directionally consistent with human bargaining literature. At the market level, prices exhibit persistent within-tick dispersion without convergence, directional but attenuated shock responses, and a demand-supply asymmetry that runs counter to classical supply-demand theory. Surplus-maximising matching improves welfare by 7.2% relative to random matching, demonstrating that mechanism design can improve aggregate outcomes independently of agent sophistication.

A consistent methodological insight runs through the programme: protocol-level design choices—including the reference price framing, the accept-condition block, and the finite-horizon counteroffer action space—shape the behaviours experiments can observe. All findings should therefore be understood as characterising a specific model-plus-prompt-plus-protocol combination, not LLM negotiation capability in general.

8.2 Implications

The findings carry implications for four audiences.

For understanding AI negotiation behaviour. Under the tested conditions, LLM agents reproduce the directional qualitative signatures of deadline pressure, Boulware concession, and first-offer anchoring documented in human negotiation studies. These alignments suggest viability for testing protocol designs and studying how informational environments shape bargaining outcomes.

However, systematic departures—price stickiness and the absence of threat-of-rejection dynamics under the counteroffer protocol—limit the extent to which LLM agents mimic the full range of human negotiation behaviour. Direct human comparison under identical protocol is required before stronger claims can be made.

For experimental methodology. Simulator design choices—feasibility enforcement, prompt steering, reference price framing—can mask or distort the behaviours experiments are designed to study. The gate-off design was motivated precisely by preliminary observations that gate-on runs suppressed deadline effects and concession trajectories; the Experiment 4 extension provides a parallel lesson for reference price framing. Researchers using LLM agents in structured settings should treat such interventions as controllable variables, report their involvement rates, and conduct ablation studies to separate agent capability from system-level enforcement.

For automated negotiation deployment. LLM agents reach agreement in the majority of sessions when the ZOPA is wide (overall deal rate: 78.6%), but price stickiness limits their usefulness as price-discovery mechanisms. Deployed systems using similar prompt architectures may exhibit systematic mis-pricing that does not self-correct as market conditions change, because agents anchor to the reference price embedded in the prompt rather than adapting to new fundamentals. The scope of this concern beyond the tested model-prompt-protocol combination is unknown.

For economic theory. Several patterns—price dispersion, first-mover advantage at the micro level, and directional shock response—are directionally consistent with bilateral bargaining theory, even though agents lack sophisticated strategic reasoning. The current experiments cannot determine whether these patterns arise from the bargaining structure, model-specific training distributions, or prompt architecture; no experiment independently varies agent sophistication. The appropriate interpretation is modest: qualitative predictions of bilateral bargaining theory can emerge in a simulator with unsophisticated agents, not that LLM agents are strategically rational or constitute valid substitutes for human participants in experimental economics.

8.3 Future Work

Several directions would strengthen and extend the current findings.

Prompt ablation. Prompt-level ablation—removing the accept-condition block, removing the monotonic concession instruction, extending the action space with a voluntary walk-away option, and varying the deadline-salience language—is needed to clarify the boundary between agent-driven behaviour and protocol-induced compliance. the Experiment 4 extension demonstrates that reference

price framing is a substantial source of price-level variation; similar ablations for other prompt components would clarify how much of the observed bargaining behaviour reflects genuine LLM decision-making. The platform already supports prompt configuration via YAML, making these ablations straightforward to execute.

Multi-model comparison. All reported experiments use a single model (Qwen2.5-14B-Instruct). Repeating the experimental configuration with models of different sizes (7B, 70B) and architectures under identical prompt and gate settings would test whether price stickiness, asymmetric concession burden, and deadline sensitivity are model-scale phenomena or general properties of LLM bargaining. This is the highest remaining external-validity gap.

Gate ablation. A within-configuration gate ablation—comparing gate-on and gate-off under the same model—would provide controlled evidence on how feasibility enforcement affects observed outcomes. This would complement the preliminary observation that gate-on runs suppressed behavioural signals.

Extended market replication. Experiments 3 and 4 were replicated across three seeds, which improves confidence in the directional findings but reveals substantial cross-seed variance (notably in Experiment 4’s shock response trajectories). Additional seeds and longer tick horizons (beyond 10 ticks) would help distinguish persistent market-level patterns from seed-specific fluctuations.

Supply-demand mechanism isolation. Experiment 6 establishes the asymmetric price response to symmetric cost and value shifts but cannot isolate the responsible mechanism. Three targeted ablations would address this: (i) varying the size of the demand shift to test whether the price response is linear or threshold-gated; (ii) running the supply-shock condition with different first-mover protocols (seller-first or simultaneous) to test whether cost-floor pass-through depends on the buyer-first turn order; (iii) testing with and without the reference price signal to isolate anchoring effects.

Reputation and communication experiments. The platform supports two additional experiment families—reputation and repeated interaction (using the round-robin matcher and reputation agent) and communication strategy ablation (using prompt-level communication modifiers)—for which infrastructure is implemented but empirical evaluation has not been conducted. Both require the gate to be disabled so that strategic adaptation can manifest over multiple rounds.

Statistical analysis. The post-hoc anchor-mode comparison (Section 6.3.3) represents a first step toward formal statistical evaluation, but most findings in this thesis remain at the descriptive level. Broader application of hypothesis tests, confidence intervals, and effect-size measures—together with additional

seeds to increase independent replication—would strengthen the evidential basis of the current findings.

Human–LLM comparison. Running the same experimental conditions with human participants would enable direct comparison of bargaining behaviour. This would clarify whether the observed patterns are specific to LLMs or whether they extend to any agent operating with explicit constraint information.

Stronger baselines. The rule-based agent provides a deterministic baseline but follows a simple linear concession strategy. Adding stronger algorithmic baselines—such as Nash-equilibrium-computing agents or game-theoretic optimal strategies—would help calibrate the gap between LLM behaviour and theoretically optimal play.

The platform is a testbed, not a finished product. Its value lies in enabling structured, logged comparison of LLM bargaining behaviour against theoretical reference points and documented human negotiation patterns—and in making simulator design choices, such as the informational anchor environment and prompt architecture, explicit and configurable variables rather than invisible background conditions.

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